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### Vehicle

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### Abstract

A suspension assembly for a vehicle includes a lower alignment arm, an upper alignment arm, and a linear force element. The lower alignment arm includes a first portion, second portion, a third portion, and a linear force element mount. The linear force element mount includes a bearing for pivotably coupling the linear force element to the lower alignment arm.

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## Background/Summary

(1) The present application is a continuation-in-part of U.S. patent application Ser. No. 16/861,859, filed Apr. 29, 2020, and entitled "VEHICLE," which claims priority to U.S. Provisional Patent Application Ser. No. 62/840,654, filed Apr. 30, 2019, and entitled "VEHICLE," the complete disclosures of which are expressly incorporated by reference herein.

## FIELD OF THE DISCLOSURE

(1) The present invention relates to side-by-side vehicles and all-terrain vehicles.

## BACKGROUND OF THE DISCLOSURE

(2) Generally, all-terrain vehicles ("ATVs") and utility vehicles ("UVs") are used to carry one or two passengers and a small amount of cargo over a variety of terrains. Due to increasing recreational interest in ATVs, specialty ATVs, such as those used for trail riding, racing, and cargo hauling have entered the market place. Most ATVs include seating for up to two passengers which are either seated side-by-side or with the passenger positioned behind the driver of the ATV. Side-by-side ATVs, in which the driver and passenger are seated beside each other on laterally spaced apart seats, have become popular because of the ability to allow the passenger to share the driver's viewpoint.

## SUMMARY OF THE DISCLOSURE

(3) In one embodiment of the present disclosure, a suspension arm for a vehicle comprises a first portion and a second portion coupled to the first portion. The first portion and the second portion together define an upper surface of the suspension arm.

(4) In a further embodiment of the present disclosure, a suspension assembly for a vehicle comprising an alignment arm comprising a mounting assembly having a bearing plate and a bearing member. The bearing member is removably coupled within a portion of the bearing plate. The suspension assembly further comprising a linear force element movably coupled to the alignment arm at the bearing member.

(5) In another embodiment of the present disclosure, a suspension arm for a vehicle comprising an upper portion and a lower portion. An upper extent of the lower portion overlaps a lower extent of the upper portion to define overlapping sides of the upper and lower portions. The upper portion is coupled to the lower portion at the overlapping sides.

(6) In yet another embodiment of the present disclosure, a method of assembling a suspension arm for a vehicle, the method comprising providing a first portion, providing a second portion, providing a third portion, permanently coupling the first portion to the second portion along an upper surface of the first portion, and permanently coupling the first portion to the third portion along a lower extent of the first portion.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front left perspective view of the vehicle of the present invention without the body panels;

(2) FIG. 2 is a right rear perspective view of the vehicle of FIG. 1;

(3) FIG. 3 is a left side view of the vehicle of FIG. 1;

(4) FIG. 4 is a right side view of the vehicle of FIG. 1;

(5) FIG. 5 is a top view of the vehicle of FIG. 1;

(6) FIG. 6 is a front view of the vehicle of FIG. 1;

(7) FIG. 7 is a rear view of the vehicle of FIG. 1;

(8) FIG. 8 is a front left perspective view of the frame of the vehicle of FIG. 1;

(9) FIG. 9 is a rear right perspective view of the frame shown in FIG. 8;

(10) FIG. 10 is a front left perspective view of the front suspension of the vehicle;

(11) FIG. 11 is a left side view of the front suspension of the vehicle;

(12) FIG. 12 is a view similar to that of FIG. 10, having removed the frame;

(13) FIG. 13 is a rear perspective view of the suspension as shown in FIG. 12;

(14) FIG. 14 is an enlarged view of the front right side suspension;

(15) FIG. 15 is a partially exploded view of the suspension of FIG. 14;

(16) FIG. 16 shows a floating rotor design for the vehicle;

(17) FIG. 17 is a left front perspective view of the front removable frame portion;

(18) FIG. **18** is a partially exploded view of the front removable frame portion of FIG. **19**;  
(19) FIG. **19** is a front left perspective view of the final drive removed from the front frame of FIG. **18**;  
(20) FIG. **20** is a bottom view of the vehicle showing the rear suspension;  
(21) FIG. **21** is a right front perspective view of a portion of the rear suspension;  
(22) FIG. **22** shows an exploded view of the rear suspension of FIG. **21**;  
(23) FIG. **23** is a front left perspective view of the powertrain of the vehicle;  
(24) FIG. **24** is a right rear perspective view of the engine;  
(25) FIG. **25** is a right rear perspective view of the engine oil pan and lubrication system;  
(26) FIG. **26** is a view similar to that of FIG. **25** showing the front scavenge pump exploded away from the oil pan;  
(27) FIG. **27** is a cross-sectional view through lines **27-27** of FIG. **25**;  
(28) FIG. **28** is a cross-sectional view through lines **28-28** of FIG. **25**;  
(29) FIG. **29** is a left-hand side view of the engine mounts;  
(30) FIG. **30** is a left rear perspective view of the engine mounts;  
(31) FIG. **31** is similar to that of FIG. **30**, showing the engine and engine mounts in an exploded manner away from the frame;  
(32) FIG. **32** is a right-hand side view of the engine mounts;  
(33) FIG. **33** is a right rear perspective view of the engine mounts;  
(34) FIG. **34** is a left front perspective view of the transmission coupled to the front final drive;  
(35) FIG. **35** is a rear perspective view of the transmission shown in FIG. **34**;  
(36) FIG. **36** is a cross-sectional view through lines **36-36** of FIG. **34**;  
(37) FIG. **37** is an exploded view of the slip clutch portion of the transmission;  
(38) FIG. **38** shows a diagrammatical flow chart showing the torque control model;  
(39) FIG. **39** is a front perspective view of the continuously variable transmission (CVT) of the powertrain shown in FIG. **23**;  
(40) FIG. **40** is the inner housing of the CVT shown with the drive and driven clutches removed;  
(41) FIG. **41** is a cross-sectional view through lines **41-41** of FIG. **39**;  
(42) FIG. **42** shows a coolant level switch for the coolant bottle;  
(43) FIG. **43** is a front perspective view of the utility bed of the vehicle;  
(44) FIG. **44** shows a right-side view of the passenger seat and fuel tank configuration;  
(45) FIG. **45** shows a top view of the seat shown in FIG. **44**;  
(46) FIG. **46** shows an enlarged view of the rear cab frame support area;  
(47) FIG. **47** is a fragmentary enlargement of the right rear portion of the cab frame support area shown in FIG. **46**;  
(48) FIG. **48** is a cross-sectional view through lines **48-48** of FIG. **46**;  
(49) FIG. **49** is a view showing the fully assembled taper lock fitting;  
(50) FIG. **50** shows an enlarged portion of the area denoted in FIG. **46**;  
(51) FIG. **51** shows a left front perspective view of an alternative front frame portion to that of the front removable frame portion of FIG. **17**;  
(52) FIG. **52** is a partially exploded view of the front frame portion of FIG. **51**;  
(53) FIG. **53** is a front left perspective view of the final drive removed from the front frame portion of FIG. **51**;  
(54) FIG. **54** is a left side view of front frame portion of FIG. **53**;  
(55) FIG. **55** is a right side view of front frame portion of FIG. **53**;  
(56) FIG. **56** is a left front perspective view of an alternative frame of the vehicle of FIG. **1**;  
(57) FIG. **56A** is a left front perspective view of a portion of the frame of FIG. **56**;  
(58) FIG. **57** is a right rear perspective view of the frame of FIG. **56**;  
(59) FIG. **58** is a left front perspective view of the frame of FIG. **56** coupled to an alternative cab frame;

(60) FIG. **59** is a detailed perspective view of a tie-down member of the frame of FIG. **56**;

(61) FIG. **60** is a detailed perspective view of a tie-down member of the frame of FIG. **57**;

(62) FIG. **61** is a left front perspective view of the cab frame of FIG. **58**;

(63) FIG. **62** is a left rear perspective view of an underside of the cab frame of FIG. **61**;

(64) FIG. **63** is a left rear perspective view of a support area of a rear portion of the frame of FIG. **57** configured to support a utility bed, a portion of an air intake, a portion of an exhaust, and a portion of a cooling system;

(65) FIG. **64** is a left rear perspective view of the support area of FIG. **63**;

(66) FIG. **65** is an exploded view of the support area of FIG. **64**;

(67) FIG. **66** is a left rear perspective view of an underside of the support area of FIG. **63**;

(68) FIG. **67** is a right rear perspective view of the underside of the support area of FIG. **66**;

(69) FIG. **68** is an exploded view of the support area of FIG. **67**;

(70) FIG. **69** is a left front perspective view of a portion of an alternative front suspension of the vehicle of FIG. **1**;

(71) FIG. **70** is an exploded view of a lower portion of a linear force element of the front suspension of FIG. **69** and a clevis of the linear force element;

(72) FIG. **71** is a right rear perspective view of an alternative front braking system of the vehicle of FIG. **1**;

(73) FIG. **72** is a left front perspective view of a brake disc and brake caliper of the front braking system of FIG. **71**;

(74) FIG. **73** is an exploded view of the brake caliper and brake pads of the front braking system of FIG. **72**;

(75) FIG. **74** is a right rear perspective view of an alternative rear braking system of the vehicle of FIG. **1**;

(76) FIG. **75** is a left front perspective view of an alternative rear suspension of the vehicle of FIG. **1**;

(77) FIG. **76A** is a left front perspective view of a toe link coupling of the rear suspension of FIG. **75**;

(78) FIG. **76B** is an exploded view of the toe link coupling of FIG. **76A**;

(79) FIG. **76C** is a cross-sectional view of the toe link coupling of FIG. **76A**, taken along line **76C-76C** of FIG. **76A**;

(80) FIG. **77A** is a left front perspective view of a trailing arm of the rear suspension of FIG. **75**;

(81) FIG. **77B** is an exploded view of the trailing arm of FIG. **77A**;

(82) FIG. **78** is a cross-sectional view of the trailing arm of FIG. **77A** taking along line **78-78** of FIG. **77A**;

(83) FIG. **79** is a perspective view of an arc spring assembly of the powertrain of the vehicle of FIG. **1**;

(84) FIG. **80** is a perspective view of the arc spring assembly coupled to a crankshaft of the engine of the powertrain of FIG. **79**;

(85) FIG. **81** is an exploded view of the arc spring assembly of FIG. **79**;

(86) FIG. **82** is a perspective view of an arc spring damper assembly of the arc spring assembly of FIG. **79**;

(87) FIG. **83** is a cross-sectional view of the arc spring damper assembly of FIG. **82**, taken along line **83-83** of FIG. **82**;

(88) FIG. **84** is a cross-sectional view of the arc spring damper assembly of FIG. **82**, taken along line **84-84** of FIG. **82**;

(89) FIG. **85** is a left front perspective view of a front drive and mounting assembly of the frame of FIG. **56**;

(90) FIG. **86A** is an exploded view of the front drive and mounting assembly of FIG. **85**;

(91) FIG. **86B** is a right rear perspective view of the front drive and mounting assembly of FIG. **85**;

(92) FIG. **87A** is a left front perspective view of a rear drive and mounting assembly of the frame of FIG. **57**;

(93) FIG. **87B** is an exploded view of the rear drive and mounting assembly of FIG. **87A**;

(94) FIG. **88** is a perspective view of a breather tube of the powertrain of the vehicle of FIG. **1**;

(95) FIG. **89** is a further perspective view of the breather tube of FIG. **88**;

(96) FIG. **90** is a cross-sectional view of a portion of the breather tube of FIG. **88**, taken along line **90-90** of FIG. **88**;

(97) FIG. **91** is an exploded view of the breather tube of FIG. **89**;

(98) FIG. **92** is a cross-sectional view of a check valve of the breather tube of FIG. **89** when the vehicle of FIG. **1** is in a tip- or roll-over condition;

(99) FIG. **93** is a cross-sectional view of the check valve of FIG. **92** when the vehicle of FIG. **1** is in an upright position on a ground surface;

(100) FIG. **94** is a right rear perspective view of air intake bezels of the vehicle of FIG. **1**;

(101) FIG. **95** is a perspective view of the air intake bezel of FIG. **94**;

(102) FIG. **96** is an exploded view of the air intake bezel and a bezel cover;

(103) FIG. **97** is a right rear perspective view of an exhaust of the vehicle of FIG. **1**;

(104) FIG. **98** is a right rear perspective view of a muffler of the exhaust of FIG. **97**;

(105) FIG. **99** is a cross-sectional view of an outlet portion of the exhaust, taken along line **99-99** of FIG. **98**;

(106) FIG. **100** is a perspective view of a coolant bottle of the vehicle of FIG. **1**;

(107) FIG. **101** is a left rear perspective of the frame of FIG. **57** and a fuel tank with a vent line;

(108) FIG. **102** is a right rear perspective view of a shifting assembly for a shiftable transmission of the vehicle of FIG. **1**;

(109) FIG. **103** is an elevational view of a portion of the shifting assembly and the shiftable transmission of FIG. **102**;

(110) FIG. **104** is a left front perspective view of a panel within a seating area of the vehicle of FIG. **1**;

(111) FIG. **105** is an exploded view of the panel of FIG. **104**;

(112) FIG. **106** is a perspective view of an air intake grille and a portion of a hood of the vehicle of FIG. **1**;

(113) FIG. **107** is an underside view of the hood and the air intake grille of FIG. **106**;

(114) FIG. **108** is an exploded view of the hood and the air intake grille of FIG. **107**;

(115) FIG. **109** is a perspective view of the air intake grille and a hinge assembly;

(116) FIG. **110** is a further perspective view of the air intake grille and a hinge assembly of FIG. **109**;

(117) FIG. **111** is another perspective view of the air intake grille and a hinge assembly of FIG. **110**;

(118) FIG. **112** is a right front perspective of a lightbar assembly coupled to a portion of the cab frame of the vehicle of FIG. **1**;

(119) FIG. **113** is a perspective of the mounting of the lightbar assembly to the cab frame of FIG. **112**;

(120) FIG. **114** is a perspective view of an underside of a mounting member of the lightbar assembly of FIG. **112**;

(121) FIG. **115** is a rear perspective view of the lower suspension arm of FIG. **12**;

(122) FIG. **116** is a front perspective view of the lower suspension arm of FIG. **115**;

(123) FIG. **117A** is an exploded view of the lower suspension arm of FIG. **116**;

(124) FIG. **117B** is a further exploded view of the lower suspension arm of FIG. **117A**;

(125) FIG. **118** is a perspective view of a ball joint housing of the lower suspension arm of FIG. **115**;

(126) FIG. **119** is a cross-sectional view of the lower suspension arm of FIG. **116**, taken along line

**119-119** of FIG. **116**;

(127) FIG. **120** is a cross-sectional view of the lower suspension arm of FIG. **116**, taken along line **120-120** of FIG. **116**;

(128) FIG. **121** is a front perspective view of the lower suspension arm of FIG. **116** coupled to the clevis for the linear force element of the suspension assembly of FIG.

(129) FIG. **122** is an exploded view of the lower suspension arm and clevis of FIG. **121**;

(130) FIG. **123** is a perspective view of the clevis is FIG. **122**;

(131) FIG. **124** is a front view of the lower suspension arm of FIG. **116** and a cross-sectional view of the clevis of FIG. **123**, showing the clevis relative to the axle or half shaft during operation of the vehicle at ride height;

(132) FIG. **125** is a front view of the lower suspension arm of FIG. **116** and a cross-sectional view of the clevis of FIG. **123**, showing the clevis relative to an axle or half shaft during operation of the vehicle at full rebound of the suspension assembly; and

(133) FIG. **126** is a perspective view of the upper suspension arm of the suspension assembly of FIG. **12**.

#### DETAILED DESCRIPTION OF THE DRAWINGS

(134) With reference first to FIGS. **1-7**, the vehicle of the present invention will be described. As shown, the vehicle is generally depicted as reference number **2** which includes front wheels **4** and rear wheels **6**. Front wheels **4** are comprised of rims **8** and tires **10**, whereas wheels **6** are comprised of rims **14** and tires **16**. Wheels **4** and **6** support a vehicle frame which is shown generally at **20** (FIG. **3**) and which supports a seating area **22** comprised of a driver's seat **24** and a passenger seat **26**. A cab frame is shown at **28** and generally extends over the seating area **22** to protect the passengers from such objects as tree branches, etc. A passenger grab bar **32** is provided for the passenger in seat **26**. As best shown in FIG. **2**, vehicle **2** further includes a steering assembly at **34** for steering front wheels **4** whereby the steering assembly **34** includes a steering wheel **36** which could be both tiltable and longitudinally movable as described in U.S. Provisional Patent Application Ser. No. 62/615,684 filed Jan. 10, 2018, the subject matter of which is disclosed herein in its entirety.

(135) Vehicle **2** further includes a front suspension at **40**, which in the present disclosure is a double A-arm suspension and further includes a rear suspension **42**, which as shown in the present disclosure is a trailing arm-type suspension. As shown best in FIGS. **2-4** (see also FIG. **22**), powertrain is comprised of engine **50** (FIG. **3**), a continuously variable transmission (CVT) **52** (FIG. **3**) and a shiftable transmission **56** (FIG. **3**) which is operated by a shifter assembly **60** (FIG. **5**). In a preferred version of the present invention, the vehicle is a four-wheel drive vehicle including a front final drive **70** (FIG. **2**) having front stub shafts **72** coupled to the front final drive **70** for driving the front wheels **4**. A rear final drive **76** (FIGS. **2, 7**) is provided to drive half shafts **78** which drives rear wheels **6**. As shown vehicle **2** may also include an outer body **80** including a hood **82**, side panels **84**, doors **86**, a utility bed **88** and rear panels **90**. The vehicle described herein may be further configured as shown in U.S. Pat. No. 8,827,028 and/or U.S. Patent application Ser. No. 62/615,684, the subject matter of which is incorporated herein by reference.

(136) With reference now to FIGS. **8-9**, frame **20** will be described in greater detail. With reference first to FIG. **8**, frame **20** is comprised of outer frame tubes **100** and inner frame tubes **102**. Outer tubes **100** include a longitudinally extending portion **104** and vertically extending portions **106**. Frame tubes **102** include longitudinally extending portions **108** and vertically upstanding frame tubes **110**. Outer frame tubes **112** extend forwardly and generally parallel with frame tubes **104** and couple to upright frame tubes **114**. Frame tubes **114** support a U-shaped frame tube **116** having couplers **118** to which cab frame **28** is attached. U-shaped tube **116** is supported by a front frame portion **120** comprised of a fixed front frame portion **122** and a removable front frame portion **124**. Fixed front frame portion **122** is comprised of three sets of upwardly extending frame tubes, namely frame tubes **126**, frame tubes **128** and frame tubes **130**. Fixed frame portion also includes

transverse frame members **132** and **134** (FIG. **9**). Finally, fixed front frame portion **122** is comprised of longitudinal tube portions **140**.

(137) With reference still to FIG. **8**, a seating support is generally shown at **150** including transverse frame tubes **152** and **154** and a longitudinal frame tube pair **156** which support the seat **24** and a frame rail **158** to support seat **26**. With reference now to FIG. **9**, a rear of the frame **20** will be described.

(138) With reference to FIG. **9**, a powertrain support area **170** is provided including rearwardly extending frame tubes **172** and upwardly extending frame tubes **174**. Frame tubes **174** extend upwardly and include forwardly projecting tube portions **174a** and vertically extending portions **174b**.

(139) With reference now to FIG. **9**, a cab frame support area **180** is provided having a transverse tube at **182**, which couples to forwardly projecting tube portions **174a** of frame tubes **174**. Transverse tube **182** is also coupled to frame tubes **106**, **184**. Taper joint brackets **186** couple to frame tube **182** as more described herein. Frame tubes **190** extends upwardly from tubes **174a** and include brackets **196**, to which a rear portion of cab frame **28** is coupled, as further described herein.

(140) With reference now to FIGS. **8** and **9**, a utility bed support area **200** will be described in greater detail. Support area **200** includes transverse tube **202** extending between tubes **106** and supported by upright frame tubes **110**. A rear transverse tube **206** extends transversely of and is coupled to frame tubes **174**. Support area **200** further includes longitudinally extending tubes **208** which couple at the rear to tube **206** and to brackets **210** at the front end thereof.

(141) With respect still to FIGS. **8** and **9**, door support brackets are provided on both the driver and passenger side, whereby a bracket **224** and **226** extend forwardly from tube **106** on the driver's side and a bracket **224** and **228** extend forwardly on the passenger side.

(142) Finally and with respect still to FIGS. **8** and **9**, engine mount brackets will be described. As shown in FIG. **8**, a left hand side mount assembly is shown at **240** and a right hand side mount assembly is shown at **242** in FIG. **9**. Left and right hand side brackets are also shown in FIG. **9** at **244** and **246**.

(143) With reference now to FIGS. **10-15**, front suspension will be described in greater detail. As shown in FIG. **10**, frame **20** includes upper couplings **260**, **262** and lower couplings **264** and **266**. Rear couplings **268** and **270** (FIG. **11**) are also provided for the left-hand side front suspension **40** and mirror image couplings are provided for the right-hand side suspension **40**. Suspension **40** includes an upper A-arm **280**, lower A-arm **282** and a linear force element (LFE) **284** which is shown as a spring over shock. Front final drive **70** is shown positioned intermediate the front couplings **260-266** where half shafts **290** extend from the front final drive **70** to drive the wheel hub **292**. Shock absorber **284** is coupled to the frame at an upper bracket **296** and is coupled to the lower A-arm **282** by way of a clevis **298** which flanks the half shaft **290** as described in greater detail herein.

(144) Front suspension **40** further comprises a stabilizer or torsion bar **300** coupled to frame tubes **128** and coupled to lower A-arms **282** as further described herein. As shown in FIGS. **12** and **13**, torsion bar **300** includes a transverse rod **302** coupled to link arms **304** which in turn are coupled to rods **306**. Rods **306** are coupled to brackets **308** positioned on the lower A-arms **282**. As shown in FIGS. **14** and **15**, bracket **282** is comprised of plate portions **310**, **312** having apertures **314**, **316** therethrough to receive fasteners **320** therethrough (FIG. **12**). As shown best in FIG. **15**, lower A-arm **282** includes a shock mount **330** having an aperture at **332**. A sleeve **334** is received into aperture **332** from one side thereof and a bearing **336** is received on the opposite side thereof and held in place with aperture **332** by way of snap ring **338**. A second sleeve **340** is positioned within aperture **332** whereupon fastener **342** may be received through sleeves **334**, bearing **336** and sleeve **340** whereupon a fastener **344** couples to fastener **342** to retain clevis **298**. As shown best in FIG. **15**, clevis **298** includes clevis legs **350** having apertures at **352** to receive fastener **342** therethrough.

The legs **350** of the clevis **298** are elongated in the vertical direction to allow the free movement of the half shafts **290** within the clevis **298** during the operating movement of the suspension. An axis **354** that runs down the axial center of the shock member intersects the half shaft **290**.

(145) With reference now to FIGS. **12-15**, each of the control arms **280, 282** are coupled to the frame by way of couplers **370, 372, 374, 376** and outer couplers **380, 382, 384** and **386**. It should be appreciated that A-arms **280** and **282** are coupled to the frame by way of couplings **260-270** on the frame as described above and as shown in FIGS. **10-11**. Outer ends of A-arms **280, 282** retain a steering spindle **390** by way of upper and lower ball joints **392, 394**. As shown best in FIG. **13**, steering spindle **390** includes a steering link **398** to which steering arm **400** is attached by ball joint **402**. Steering arms **400** are coupled at their inner ends to steering motor **410** having an input at **412** which couples through to steering system **34** so as to turn the left and right front wheels **4** through steering rack **414**. Steering spindles **390** rotate about an axis through ball joints **392, 394** and at the same time retain wheel hub **292** and front braking system **420** thereto, as described in greater detail herein.

(146) With reference now to FIG. **16**, front braking system **420** will be described in greater detail. As shown, braking system **420** is coupled to the steering spindle **390** and comprises brake caliper **422**, disc pads **424** and brake disc **426**. As shown, hub **292** includes an inner hub portion **292a** and an outer hub portion **292b**. Inner hub portion **292a** is directly coupled to half shaft **290** so as to be driven, and hub portion **292a** includes a plurality of fasteners **428**. Hub portion **292b** includes a plurality of apertures at **430** which receive fasteners **432** therethrough and disc **426** includes apertures **436**. Hub portion **292b** includes a plurality of apertures **440** which align with fasteners **428** so that hub portion **292b** and disc **426** couple directly to the hub portion **292a**. Caliper **422** and disc pads **424** couple to a brake mount at **442** such that disc pads **424** are held on opposite sides of the brake disc **426**. Alternatively, the brake disc could be shown as a single component, comprising the disk **426**, hub portion **292b** and fasteners **432**. Also, fasteners **432** may be rivets such as buck rivets.

(147) With reference now to FIGS. **17-19**, the removable front frame portion **124** will be described in greater detail. As shown in FIGS. **17** and **18**, the removable front frame portion **124** is coupled to the frame just adjacent to the suspension couplings **260, 262, 264** and **266**. Removable front frame portion **124** includes a plate section **450** to which a bracket **452** is mounted which can provide a location for a winch mount. A tow bar **454** is mounted to the bracket **452** and a hook **456** is mounted at a lower portion of plate portion **450**. Plate portion **450** includes four mounting holes at **458** which provide access to fasteners **460**. Each of the apertures **458** further includes a locator aperture **464** as further described herein.

(148) As shown, fixed frame portion **122** includes an upper frame tube portion **470** which is substantially parallel to lower frame tube **102** where each of the upper frame tubes **470** and lower frame tubes **102** include inserts **480, 482, 484** and **486**. As shown, each of the couplings **480-486** perform two functions, first to define a portion of the suspension couplings **260-266** as well as provide a mounting location for the removable frame portion **124**. More particularly, insert **480** includes a cylindrical portion **480a** to which a bracket arm **480b** is mounted having an aperture which defines a mounting location for upper A-arm **280**. At the same time, insert **480** includes a raised land **480c**, a tear-drop configuration and having a threaded aperture at **480d** and a locating lug at **480e**. In a like manner, insert **482** includes a cylindrical portion **482a**, plate **482b**, raised land **482c**, threaded aperture **482d** and locating lug **482e**.

(149) As shown in FIG. **19**, insert **484** includes a cylindrical portion **484a**, plate portion **484b**, raised land **484c**, threaded aperture **484d** and locating lug **484e**. In a like manner, insert **486** includes a cylindrical portion **486a**, plate portion **486b**, raised land **486c**, threaded aperture **486d** and locating lug **486e**. It should be recognized that each of the cylindrical portions **480a, 482a, 484a** and **486a** include a reduced diameter portion extending rearwardly therefrom which is receivable into respective frame tubes **102** or **470** whereby the inserts **480-486** may be held in place by

way of industrial adhesives or other such bonding as welding.

(150) It should be noted from FIGS. **18** and **19**, that the lands **480c-486c** provide the interface for coupling the removable frame portion **124** to the fixed frame portion **122**. Namely, each of the apertures **458** and the movable frame portion **120** align with associated threaded portions **480d-486d** and each of the alignment apertures **464** aligns with corresponding locating lugs **480e-486e**. Thus, once aligned, fasteners **460** are receivable through corresponding apertures **458** and into their corresponding threaded apertures **480d-486d**. Thus to remove the front final drive **76**, the front removable frame portion **124** is simply removed by uncoupling the fasteners **460** whereby the removable front frame portion **124** moves to the location shown in FIG. **18**, and whereby front final drive portion **76** may be removed by uncoupling fasteners **490** and **492** (FIG. **19**).

(151) It should be noted from FIG. **19** that the removal of the removable frame portion **124** and the front final drive **76** does not require the removal of the front suspensions **40** as the couplings **260-266** remain fixed to the fixed frame portion **124**. It should also be noted that the front final drive **76** includes mounting inserts **496** which couple with the fasteners **490** in a manner substantially as shown in U.S. patent application Ser. No. 15/389,147, the subject matter of which is incorporated herein by reference.

(152) In addition, the upper couplings **260, 262** serve two functions. First, each of the couplings **260, 262** provide the mounting structure for the front suspension as previously described but also include sidewalls **260a, 262a** (FIG. **19**) which include the apertures for receiving fasteners **490**.

(153) With reference now to FIGS. **20-22**, rear suspension **42** will be described in greater detail. As shown best in FIG. **20**, rear suspension **42** is shown from an underside thereof in association with the engine **50**, CVT **52** and shiftable transmission **56**. As shown, CVT **52** is positioned forward in the support area **200** and occupies a space normally taken up by a torsion bar. Rear suspension **42** includes a pair of trailing arms **500** coupled at a front end thereof to the frame (FIG. **3**) whereby the trailing arms rotate about axis **502** (FIG. **3**), radius arms **506** are coupled at an inner end **508** to frame **20** and at outer ends to the trailing arms **500**.

(154) With reference now to FIGS. **21** and **22**, the torsion bar assembly will be described in greater detail. As shown, an upper side of the trailing arm includes a bracket **512** for mounting the torsion bar assembly **510** and a bracket **514** for mounting the rear linear force element **516** (FIG. **1**). As shown, torsion bar assembly **510** couples to the same bracket **210** that mounts the frame tubes **208** (FIG. **9**), which supports the utility bed **88** (FIG. **2**). Bracket **210** includes a bracket portion **210a** for mounting tubes **208** and a bracket portion **210b** which faces rearwardly and downwardly for mounting the torsion bar assembly **510**. Torsion bar assembly **510** includes a transverse bar portion **520** to which link arms **522** are fixedly mounted. Link arms **522** couple with link rods **524** which in turn couple to brackets **512**. Torsion bar assembly **510** may be further configured as shown in U.S. Pat. No. 8,827,019, the subject matter of which is incorporated herein by reference.

(155) With reference now to FIGS. **21** and **22**, the torsion bar assembly **510** includes bearings **530** having bearing halves **532** and a bearing collar **534**. Bearing halves **532** are split to receive a reduced diameter section **540** in a rotatable manner and are received within each of the collars **534**. Each collar **534** includes an aperture at **544** which receives fasteners **546** therethrough. Fasteners **546** extend through apertures **548** of bracket portion **210b** to couple the transverse portion **520** to the frame tube **210**. Link arms **522** have a split opening at **550** which couples to ends **552** of transverse portion **520**. Split portions **550** include an aperture at **556** which receives a fastener **558** so as to clamp the split opening **550** to the portions **552**. Link arms **522** include a coupler **560** at the opposite end which corresponds with a coupling **562** of link rods **524**. Fasteners **564** are receivable through the couplers **560, 562** and is retained by a counter fastener **566**. Lower ends of link rods **524** also includes a lower coupler at **568** which receives a fastener **570** therethrough to couple the link arms **524** to the bracket **512** with a counter fastener **572**.

(156) With reference again to FIG. **20**, it should be noted that the suspension, namely the radius rods **506**, are centered about a longitudinal center line **580** of the vehicle whereas a center line of

the engine is off-set from the longitudinal center line of the vehicle at a center line **582**. Due to the off-set of the engine **50**, the drive to the front final drive **70** and rear final drive **76** is slightly angled. More particularly, as shown in FIG. **20**, a rear prop shaft **586** extends at a slightly skewed angle **588** whereas front prop shaft **590** extends at a slightly skewed angle **592**. With reference again to FIG. **11**, front prop shaft **590** includes a coupling at **594**, which may be in the form of a universal joint. Universal joint includes a front coupling at **596** which is directed upwardly to the front final drive **70**. Steering gear **414** is positioned as low as possible in the vehicle, and is positioned in the area defined by the inclusive angle between the front prop shaft **590** and the front coupling **596**. This optimizes bump steer.

(157) With reference now to FIGS. **23-28**, a front scavenge pump for engine **50** will be described in greater detail. As shown in FIG. **23**, engine **50** drives CVT **52** which in turn drives shiftable transmission **56**. Shiftable transmission **56** is coupled to prop shafts **586** and **590** to drive rear final drive **76** and front final drive **70** respectively. It should be appreciated that engine **50** is substantially as shown in U.S. patent application Ser. Nos. 15/595,224 and 15/595,209, the subject matter of which is incorporated herein by reference.

(158) As shown in FIG. **24**, engine **50** includes an engine block **600**, valve cover **602**, oil pan **604**, oil filtration system **606** and air intake system **608**. A timing chain cover **610** encloses the cam chain of the engine and also encloses an auxiliary scavenge pump **612** as shown in FIG. **25**. As shown in FIGS. **25** and **26**, pump **612** includes an integrated pump housing **614** within the oil pan **604**, including an input area **620** and a discharge area **622**. Housing portion **614** includes a circular opening at **624** to receive outer rotor **626** and an inner rotor **628** of the scavenge oil pump. A drive coupler **630** is provided which couples to the main oil pump **652** at one end by way of a rectangular drive **632** and couples to drive gear **634** at the opposite end by way of a truncated circular portion **636** which corresponds with a like opening at **638** of gear **634**. Pin **639** is positioned in aperture **630a** and is received in slot **628a** to drive gerotor **628**.

(159) A cover **640** is provided having an opening at **642** which receives a syphon screen at **644** which corresponds with opening at **620**. It should be appreciated from the previous description of the powertrain that the cam chain oil pump **612** is positioned longitudinally at a rear of the vehicle, such that in extreme inclines of the vehicle, oil floods to the cam chain area **646** which takes oil away from the main oil sump **648** (FIG. **25**). Thus as shown in FIG. **27**, the drive chain couples to gear **634**, drives gerotor **628** by way of pin **639** and drives pump shaft **650** of main oil pump **652** by way of rectangular end portion **632**. As shown in FIG. **28**, oil moves in a direction of arrows into the inlet **644** whereupon gerotor **628** moves the oil in the direction of arrows **660** and upward in the direction of arrows **662** and out of the cam chain area **646** and returning to the main sump area **648**.

(160) With reference now to FIGS. **29-33**, the mounting of engine **50** within frame **20** will be described in greater detail. As shown first in FIGS. **29-31**, left-hand side mount assembly **240** is shown coupled to engine **50**. As shown, mount assembly **240** includes a removable mount **670** (FIG. **31**) and is defined by a channel-shaped member having parallel plate portions **672** defining an upper platform at **674**. A front edge of the plate portions **672** includes apertures **676** and a rear edge defines apertures **678**. Brackets **680** and **682** define apertures **684** and **686** respectively which align with apertures **676** and **678**. Fastener **688** is receivable through apertures **676** and **684** while fastener **690** is receivable through apertures **678** and apertures **686**. Counter fasteners **692** couple with fasteners **688** and **690** to retain left-hand side mount assembly **240** to frame **20**.

(161) As shown in FIG. **31**, a mounting bracket is shown at **700** which couples to the backside of transmission **56** and to the left-hand side of engine **50** as shown in FIGS. **29** and **30**. Bracket **700** is somewhat L-shaped having a transverse leg portion at **702** and a longitudinal leg portion at **704**. Leg portion **702** includes apertures at **706** while leg portion **704** has apertures at **708**. Apertures **706** align with apertures **710** at the back side of the transmission **56** and apertures **708** align with apertures **712** (see FIG. **23**). Fasteners **714** (see FIG. **29**) are receivable through apertures **708** and

into threaded engagement with apertures **712** (see FIG. **23**). Fasteners **716** are receivable through apertures **706** and into threaded engagement with apertures **710**. An isolation mount **720** is provided to interface between brackets **670** and **700**. Mount **720** includes a lower mount portion **722** which receives fasteners **724** therethrough and which are receivable with threaded apertures **728** of bracket **670**. Mount **720** further includes an arcuate flange **730** having apertures at **732**. A complimentary arcuate flange **740** is positioned on bracket **700** having threaded apertures at **742**. Thus, fasteners **744** are receivable through apertures **732** and into threaded engagement with apertures **742**.

(162) With reference still to FIG. **31**, an isolation mount **750** is shown which is substantially similar to mount **720** having a lower mount portion at **752** and an arcuate flange at **754** having apertures **756**. The bracket **244** includes threaded apertures at **758** which receives fasteners **760** therethrough and through mounts **752** to retain mount **750** to bracket **244**. Transmission **56** also includes an arcuate flange at **770** (see also FIG. **34**) having threaded apertures at **772**. Fasteners **774** are receivable through apertures **756** and into threaded engagement with apertures **772** of transmission **56**. The location of arcuate flange **770** is further seen in FIG. **23**. It should be noticed from FIG. **30**, that bracket **670** is positioned adjacent to prop shaft **586** such that removal of bracket **670** and isolation mount **720** provides full access to prop shaft **586** for removal and/or replacement.

(163) With reference now to FIGS. **32** and **33**, the right-hand side engine mount will be described in greater detail. As shown, right-hand side mount assembly **242** has an upper platform **790** which receives an isolation mount **792** in a similar fashion to mount **720**. A bracket **794** is coupled to the right-hand side of the engine by way of fasteners **796** into apertures **798** (see FIG. **24**). Fasteners **800** couple the isolation mount to bracket **794**. Further details of isolation mounts may be seen in U.S. Pat. No. 9,873,316, the subject matter of which is incorporated herein by reference.

(164) With reference now to FIGS. **34-38**, a torque control feature of the transmission will be described in greater detail. With reference first to FIG. **34**, the transmission **56** is shown where the transmission includes a transmission housing at **820** having an engine interface at **822** (FIG. **35**) which defines a flange for the mounting of engine **50**. An intermediate shaft **824** is provided having an input at **826** and an output at **828**. The gearing for transmission **56** is located within housing portion **830** and which is driven through input shaft **832**. It should be appreciated that CVT **52** (FIG. **23**) couples between shafts **828** and **832** such that the engine **50** drives intermediate shaft **824** and shaft portion **828** drives a drive pulley within CVT **52** and a driven pulley of CVT **52** drives input shaft **832** of transmission **56**. As shown in FIG. **34**, an output shaft of transmission **56** is also provided at **840** which has a shaft portion **842** coupled to the front prop shaft **590** and a rear output shaft **844** (FIG. **35**) which couples to rear prop shaft **586** (FIG. **23**).

(165) With reference now to FIGS. **36** and **37**, a slip clutch is defined between the shaft portions **842**, **844** to limit torque through the front prop shaft **590**. As shown, slip clutch is defined by a plurality of friction discs **850** interposed between a plurality of clutch discs **852**. Friction discs **850** include a plurality of pads of friction surfaces **854** such as a carbon fiber material or graphite material to cause a frictional engagement between the friction discs **850** and the clutch discs **852**. Each of the friction discs **850** include teeth at **858** while the clutch discs **852** include splines **860**. Shaft portion **842** includes a reduced diameter portion **868** having a splined portion at **870**, an enlarged diameter portion at **872**, a fixed pressure plate at **874**, and a splined section at **876**. Shaft portion **842** further includes a threaded shaft portion **880** and a reduced diameter portion **882**.

(166) Shaft portion **844** includes a splined output shaft at **890**, an enlarged diameter portion at **892**, a disc basket **894** having a plurality of teeth **896** which match with teeth **858** of friction discs **850**. Shaft portion **844** further includes a helical gear **898** which is the input to lower shaft **840** driven through shaft **832**. Slip clutch further includes wave springs **900**, pressure nut **902**, movable pressure plate **904**, timing disc **906**, bearings **908**, **910** and seals **912**, **914**.

(167) The plurality of friction discs **850** and clutch discs **852** are slidably received over spline shaft portion **876** with the splines **860** of the clutch discs aligning with the splines of splined portion **876**.

This positions the plurality of stacked discs **850**, **852** positioned against an inner surface of fixed pressure plate **874**, as best shown in FIG. **36**. Movable pressure plate **904** is thereafter received also on splined portion **876** to be positioned against the last of the friction discs **850**. Wave springs **900** are thereafter positioned over shaft portion **880** to be positioned against pressure plate **904** and pressure nut **902** is threadably received on shaft portion **880** and torqued down to a preset torque where wave springs are pressing against pressure plate **904**, as shown best in FIG. **36**.

(168) As shown in FIG. **36**, an idler shaft **920** provides a helical gear **922** in engagement with helical gear **898**. Shaft **920** is drivingly coupled to input shaft **832** such that input torque from the engine is transferred through CVT **52** to transmission **56** and to outputs **842** and **844**. When the torque at **844** reaches a preset max torque, however, that is the torque where the friction discs **850** and clutch discs **852** begin to slip relative to each other, the torque through shaft **842** is limited by the slip clutch defined through discs **850**, **852**. It should be noticed that as the friction discs **850** are rotationally fixed relative to the shaft portion **844** and the clutch discs **852** are rotationally fixed relative to shaft portion **842**. When the torque at output shaft **844** is below the threshold torque, the entire shaft **840** rotates as a single piece. However, when the torque exceeds the predetermined maximum torque, the discs **850**, **852** slip relative to each other, allowing a different rotational speed between shaft portion **842** and shaft portion **844**. With reference again to FIG. **36**, the rotational speed of shaft portion **842** is measured through timing gear **906**, where timing gear **906** includes a plurality of timing tines **930** and a timing sensor is positioned in aperture **932** (FIG. **37**) to count the rotations of the tines to calculate a speed of shaft portion **842**.

(169) Thus with reference to FIG. **38**, a torque model can be provided through a plurality of sensed perimeters, calculated perimeters and output perimeters, where the shaft speed of the front shaft portion **842** is determined at **940**, the shaft speed of rear shaft portion **844** is determined at block **942** and a differential between the two shaft portions **842**, **844** is determined at block **944** so as to define a shaft speed differential at **948**. Thus, as mentioned above, if the torque at shaft portion **844** is below the predetermined maximum torque, the shaft speed differential at block **948** is 0. When the torque at shaft portion **844** exceeds the predetermined maximum torque, the discs **850**, **852** begin to slip such that a shaft speed differential is calculated, is input to the torque model at block **950** and the engine output torque is limited by the engine control unit at block **952**.

(170) With reference now to FIGS. **39-41**, CVT **52** will be described in greater detail. As shown, CVT **52** includes an inner housing **970** (adjacent the engine **50** and transmission **56**) and an outer housing **972**. CVT **52** would also include a drive side **974** and a driven side **976**, and including a drive clutch **978** and a driven clutch **980** (FIG. **41**). CVT **52** has a forward intake **982**, a rear intake **984** and an exhaust **986**. CVT **52** may also be configured as shown in U.S. Patent application Ser. No. 62/644,717, filed Mar. 19, 2018, the subject matter of which is incorporated herein by reference.

(171) As the engine **50** is not coupled directly to the CVT **52**, but rather to the engine interface **822** (FIG. **35**), the intermediate shaft **824** output **828** extends through a shaft housing **990** (FIG. **34**) which has a small profile in the radial direction. The shaft housing **990** also protrudes through the opening **992** as best seen in FIGS. **40** and **41**, providing a large space **994** around the inner sheave portion **996**. The spacing **994** provides an air volume and allows a substantial amount of air flow around the inner sheave portion **996** and outer sheave portion **998** for cooling.

(172) With reference now to FIG. **42**, a coolant bottle is shown at **1000** having an integrated switch to determine coolant level.

(173) With reference now to FIG. **43**, the utility bed **88** is shown to include channels **1002** and drains **1004** to remove contaminants so as to not drain on hot portions of the power train.

(174) With reference now to FIGS. **44** and **45**, fuel tank **1050** is shown positioned beneath and rearward of passenger seat **26**. As shown, fuel tank **1050** includes a longitudinally extending section at **1052** which is lower than and under seat **26**. Longitudinal section **1052** includes fuel pump **1054** which draws fuel from the lowest portion of the fuel tank **1050**. Fuel tank **1050** also

includes a vertically extending section at **1060** which includes the filler tube at **1062**. As shown, vertically extending section **1060** is rearward of passenger seat **26**.

(175) With reference now to FIGS. **1-5** and **46-49**, cab frame **28** will be described in greater detail. As shown first in FIGS. **1-5**, cab frame **28** generally includes longitudinally extending frame members **1070** and **1072** which couple to couplers **1074** and **1076** at a front end thereof and to couplers (not shown) at the rear thereof. A mid frame portion **1080** and **1082** extends downwardly at a position adjacent to seats **24** and **26** and couple to couplers **1086**. With reference now to FIGS. **46-49**, couplers **1086** will be described in greater detail. As shown in FIG. **47**, coupling **1086** includes a tapered lower portion at **1090** which extends along an axis **1092** which is common with the axis to coupling **186**. An upper portion of coupling **1086** includes a collar **1094** which is skewed relative to axis **1092** and includes an opening at **1096** which allows access to an opening **1100** extending through tapered portion **1090** (see FIG. **48**).

(176) Coupling **186** includes a tapered portion **1102** conforming to the tapered portion **1090** and which has a lower portion **1106** having an opening at **1108**. An aperture **1110** provides access between opening **1108** and tapered portion **1100** for receiving a fastener such as **1114**. A counter fastener **1116** is positionable through a retaining portion **1112** in opening **1096** which retains fastener **116** and prevents rotation of fastener **116**. Thus, coupling portion **1086** may be coupled to coupling **186** on tube **182** by positioning tapered portion **1090** within coupling **186**. This positions the tapered portion **1090** within the tapered portion **1102** of coupling **186** as shown best in FIG. **48**. Fastener **1114** is thereby positioned through opening **1110** which positions the threaded portion within the opening **1096** of coupling **1086**. Fastener **1116** may then be threaded onto the threads of fastener **1114** to secure the tube **1082** to tube **182**. Alternatively, brackets **196** (FIG. **9**) can also be replaced with tapered couplings **186**. Preferably, axes **1092** of all tapered couplings **186** would be parallel.

(177) With reference now to FIG. **50**, a joint **1120** is shown which provides a joinder of tubes **106** and **176** to the larger tube **182**. In this case, the tube **106** (referred to as a poke yoke pass through tube) is inserted through both walls of larger diameter tube **182**, that is, through openings **1122** and **1124**. This is a very efficient way to transmit bending moments and load the section properly, particularly where multiple tubes share a common node in a space frame welded structure.

(178) Instead of un-sharing multiple tube nodes to keep proper weld shelf, a poke yoke pass through tube profile may be added to the laser cut tube to obtain proper weld shelf, increase fatigue life, and eliminate weld fixture loading sequence issues. There is also a reduction in total weld length required as a small cost benefit. The plastics packaging required the tube to only have ~270 degrees of wrap. As also shown, the partially inserted tube **176** finds a natural stop against the poked through tube **106**.

(179) With reference now to FIGS. **51-55**, an alternate front frame will be described which could be used in place of the front frame as described in FIGS. **17-19**. As shown, front frame **1320** includes a front frame portion **1322** and a removable front bumper portion **1324** (FIG. **52**). As shown in FIG. **52**, front frame **1322** is generally comprised of lower longitudinal tube portions **1330**, upper longitudinal tube portions **1332**, front diaphragm plate **1334**, and left and right-side plates **1336** and **1338** (FIG. **53**), respectively.

(180) As shown best in FIG. **53**, lower longitudinal tube portions **1330** include threaded inserts **1340** whereas upper longitudinal tube portions **1332** include threaded inserts **1342**. Front pedestal plate **1334** is coupled to the lower longitudinal tube portions **1340** and upper longitudinal tube portions **1342** and includes an opening **1350** generally conforming to allow the passage of front drive **70** therein, as will be described in greater detail herein. Side plate **1336** is generally coupled to the left-hand side lower longitudinal tube portion **1330** and upper longitudinal tube portion **1332**, as best shown in FIG. **54**. Left plate portion **1336** includes an opening at **1360** to allow access between front drive **70** and left front wheel. Left side plate **1336** further includes a front opening at **1362** and a rearward opening at **1364** which align with mounting inserts **496** (FIG. **53**) as further

described herein. A front pedestal **1370** is positioned below aperture **1362** and a rearward pedestal **1372** is positioned below opening **1364**. Pedestal **1370** defines a mounting surface at **1374** while pedestal **1372** defines a mounting surface at **1376**.

(181) With reference now to FIG. 55, right-hand side plate **1338** is coupled to lower longitudinal tube portions **1330**, and upper longitudinal tube portions **1332**. Plate **1338** includes an opening at **1380** and which is substantially identical to the opening **1370**, and which allows access between right front wheel and final drive **70**. Right plate **1338** also includes a forward aperture at **1382** and a rearward aperture at **1384**. Note that apertures **1382** and **1384** align with, and may be viewed through, the left-hand side plate **1336** through openings **1362** and **1364**, respectively, as viewed in FIG. 54.

(182) With reference again to FIG. 53, front final drive **70** is insertable through opening **1350** through diaphragm plate **1334**. The front final drive **70** is inserted until the mounting inserts **496** align with respective openings **1362** and **1364** (FIG. 54) of the left-hand side plate **1336**. Mounting anchors **1390** and **1392** are provided, which are L-shaped in configuration having vertical portions **1390a**, **1392a**, horizontal portions **1390b** and **1392b**. Apertures **1390c**, **1392c** and **1390d** and **1392d** are provided for mounting as described herein. Anchors **1390** and **1392** align with mounting inserts **496** of front final drive **70** whereupon vertical portions **1390a** and **1392a** couple through openings **1362** and **1364** to contact mounting inserts **496**. This positions the horizontal portions **1390b** and **1392b** upon pedestals **1374** and **1376**, respectively. This also aligns apertures **1390d**, **1392d** with apertures **1374a** and **1376a**, respectively. Vertical fasteners **1400** and **1402** are receivable through apertures **1390d** and **1392d**, and apertures **1374a** and **1376a** to receive counter fasteners **1404** and **1406**. At the same time, fastener **1410** is receivable through aperture **1390c**, through the front mounting insert **496** and through aperture **1382** of right-hand side plate **1338** to receive a counter fastener **1414**. In a like manner, fastener **1412** is receivable through aperture **1392c**, rear mounting insert **496**, through aperture **1384** to receive counter fastener **1416**.

(183) In the preferred method of installation, the horizontal fasteners **1410** and **1412** are inserted first while the vertical fasteners **1400**, **1402** and anchors **1390**, **1392** are loosely installed. The horizontal fasteners **1410** and **1412** are torqued to their nominal torque rating which provides no pre-load stress to the anchors **1390** and **1392**. Rather, the anchors **1390** and **1392** remain resting upon their corresponding surfaces **1374** and **1376**. Once the horizontal fasteners **1410** and **1412** are torqued, fasteners **1400** and **1402** may then be torqued to provide the final locating position of the front final drive **70**.

(184) With reference now to FIG. 52, removable bumper **1324** includes a plate **1430** to which bumper bar **1432** is attached. Fasteners **1434** are receivable through the plate portion **1430** and into threaded fasteners within couplers **1342**. Thus, as should be appreciated, to remove front final drive **70**, bumper **1324** is removed and front final drive **70** may be pulled through opening **1350** of **1334**.

(185) Referring to FIGS. 56-114, various alternative embodiments to those disclosed herein with respect to FIGS. 1-55 and/or additional features and systems of vehicle **2** are disclosed, with like components have the same or similar reference numbers. With respect to FIGS. 56-68, an alternative embodiment of frame **20** and cab frame **28** (FIGS. 1-9) are disclosed as frame **2020** and cab frame **2028**. More particularly, as shown in FIGS. 56-60, frame **2020** includes frame tubes **2102** extending generally longitudinally and frame tubes **2114** extending generally vertically. Frame **2020** further includes frame members **4000** coupled to frame tubes **2114** and extending generally inward therefrom. Illustratively, frame **2020** includes at least two frame members **4000** and each includes a bracket **4002** coupled thereto. In various embodiments, bracket **4002** is integrally coupled with frame member **4000** and may be integrally formed with frame member **4000** (e.g., through welds, adhesive, or rivets). Bracket **4002** includes a tie-down member **4004** which, illustratively, defines a bar or other structure configured to support a load coupled thereto. In this way, bracket **4002** and tie-down member **4004** define an integrated tie-down assembly supported by frame **2020** such that a load at tie-down member **4004** and bracket **4002** also is

supported by frame **2020** and may be distributed through at least a portion of frame **2020**.

(186) Referring still to FIGS. **56-60**, frame **2020** may include additional brackets **4008** and tie-down members **4010**. For example, bracket **4009** may include a tie-down member similar to tie-down member **4004** or, as disclosed further herein, similar to a tie-down member **4010**.

Additionally, a rear portion of frame **2020** includes frame tubes **2172** extending generally longitudinally and frame tubes **2240** and **4006** extending generally vertically and coupled with and/or positioned adjacent frame tubes **2172**. Illustratively, frame tubes **4006** are positioned longitudinally rearward of frame tubes **2240** and are coupled directly with frame tubes **2172**. In one embodiment, frame **2020** includes at least two frame tubes **4006** and each frame tube **4006** includes bracket **4008** integrally coupled thereto. Bracket **4008** also is integrally coupled with frame tube **2174**, which is positioned above frame tubes **2240** and **4006** and extends generally longitudinally. As shown in at least FIG. **57**, an upper portion **4012** of bracket **4008** is integrally coupled with frame tube **2174** and a lower portion **4014** of bracket **4008** is integrally coupled with frame tube **4006**.

(187) Bracket **4008** includes tie-down member **4010** which, illustratively, defines a bar or other structure configured to support a load coupled thereto. In one embodiment, tie-down member **4010** is coupled to lower portion **4014** of bracket **4008**, however, tie-down member **4010** can be coupled to any portion of bracket **4008**. In this way, bracket **4008** and tie-down member **4010** define an integrated tie-down assembly supported by frame **2020** such that a load at tie-down member **4010** and bracket **4008** also is supported by frame **2020** and may be distributed through at least a portion of frame **2020**.

(188) It may be appreciated that brackets **4002**, **4008** and tie-down members **4004**, **4010** may be positioned at any height along frame **2020**. Additionally, brackets **4002**, **4008** and tie-down members **4004**, **4010** may be positioned within a predetermined lateral distance of a longitudinal centerline **L** (FIG. **5**) of vehicle **2**. For example, tie-down members **4004**, **4010** may be positioned laterally outboard of centerline **L** by approximately 12-36 inches and, more particularly, may be positioned laterally outboard of centerline **L** by approximately 24 inches. Further, as shown, both the left and right sides of vehicle **2** include tie-down members **4004**, **4010** such that there is at least one tie-down member **4004**, **4010** on each lateral side of centerline **L**. Additionally, tie-down members **4004**, **4010** may include a double-shear clevis attachment for loading the corresponding frame tubes and distributing the load to these sections or tubes or frame **2020**.

(189) Referring to FIGS. **61** and **62**, alternative embodiment cab frame **2028** (compared to that shown in FIGS. **1-9** and **46-50**) is shown. Cab frame **2028** extends over seats **24**, **26** (FIG. **1**) and is coupled to frame **2020**. Cab frame **2028** includes frame members or frame portions **3070**, **3072**, **3080**, **3082**, each of which is coupled to frame **2020** (FIG. **56**) through complementary couplers. In one embodiment, cab frame **2028** is a single weldment such that frame members **3070**, **3072**, **3080**, **3082**, and any other frame member of cab frame **2028** are integrally formed together (e.g., define a single weldment). Various frame members or portions of cab frame **2028** may include openings or apertures **4016** configured to expose an inner volume of such frame members or portions. Openings **4016** are illustratively shown on frame portions **3072** and **3082**, however, openings **4016** may be positioned on frame members or portions **3070** and/or **3080**. Openings **4016** are configured as pass-through openings to allow various wires, conduits, lines, or other components of vehicle **2** to extend within cab frame **2028** rather than being exposed such that these components may extend between frame members **3072** and frame portions **3080**, **3082** without being exposed to seating area **22** (FIG. **1**). This may protect such components from damage or wear and also may increase the aesthetics of vehicle **2**.

(190) Referring to FIGS. **63-68**, the rear portion of frame **2020** includes alternative embodiment utility bed support area **2200** (compared to support area **200** of at least FIGS. **8** and **9**). Support area **2200** includes tubes **2206** and **2208**, where tubes **2208** extend generally longitudinally and are coupled with tube **2206**, which extends generally laterally and perpendicularly to tubes **2208**.

Support area **2200** further includes braces **4017** extending between tubes **2208** and generally parallel to tube **2206**. Illustratively, braces **4017** are positioned longitudinally forward of tube **2206**. (191) In one embodiment, tubes **2208** are removably coupled to brackets **4020** on a cross-member **4018** with fasteners **4022**. Cross-member **4018** may be part of frame **2020** and coupled to frame tubes **2240**. As shown in at least FIGS. **64** and **65**, cross-member **4018** is positioned vertically lower than transverse tube **2182**. Tubes **2208** extend longitudinally from cross-member **4018** and a rear end of tubes **2208** is coupled with tube **2206**.

(192) As shown best in FIG. **56A**, brackets **4020** are coupled to cross-member **4018** and, more particularly, brackets **4020** extend through a recess or opening **4019** of cross-member **4018**. Illustratively, opening **4019** is a pass-through opening in which a portion of bracket **4020** extends completely through opening **4019**. In this way, a forward portion of bracket **4020** is positioned forward of cross-member **4018** and a rearward portion of bracket **4020** is positioned rearward of cross-member **4018**. Bracket **4020** may be welded or otherwise permanently fixed to cross-member **4018** at opening **2019** such that the welded or fixed joint therebetween fully surrounds opening **4019** and increases the size of the welded or fixed joint. Alternatively, the welded or fixed joint may be positioned at a portion of opening **4019** and may not fully surround opening **2019**. This configuration of brackets **4020** and cross-member **4018** allows for support area **2200** to be able to support multiple components of vehicle **2**, such as coolant bottle **3000**, utility bed **2088**, and portions of air intake assembly **2608** and exhaust **2986**, as disclosed further herein, because the load from such components may be distributed through other portions of frame **2020** (e.g., cross-member **4018**) and does not need to be supported only at brackets **4020** and tubes **2208**.

(193) Tube **2206** is removably coupled to brackets **4008** with fasteners **4024**. As shown best in FIGS. **63-65**, upper surfaces of tubes **2206** and **2208** and braces **4017** are flush with each other such that tubes **2206**, **2208** and braces **4017** define a generally planar or flat surface. In this way, support area **2200** is configured to support at least one component of vehicle **2** thereon and, more particularly, is configured to support utility bed **2088** thereon.

(194) Referring to FIGS. **67** and **68**, a bottom surface **4026** of utility bed **2088** includes guides **4028**. Guides **4028** extend downwardly from bottom surface **4026** and may be removably coupled to bottom surface **4026** or may be integrally formed with bottom surface **4026**. Guides **4028** are configured to extend along a laterally inner surface **2209** of tubes **2208** such that guides **4028** laterally overlap a portion of tubes **2208**. In this way, when utility bed **2208** is positioned on support area **2200**, guides **4028** are positioned along and in contact with inner surface **2209** of tubes **2208** such that utility bed **2088** is in the correct position for centering on and securing to support area **2200**. It may be appreciated that guides **4028** are configured to be spaced apart from each other, where necessary, to receive or otherwise accommodate components of support area **2200**, such as braces **4017**. When correctly positioned on support area **2200**, utility bed **2088** may be fastened thereto with couplers, such as bolts, screws, ties, etc.

(195) Referring still to FIGS. **63-68**, support area **2200** not only supports utility bed **2088** but also is configured to support additional components of vehicle **2**. For example, as shown best in FIGS. **63** and **66**, support area **2200** is configured to support a portion of air intake assembly **2608**, a portion of exhaust assembly **2986**, a portion of the cooling assembly of vehicle **2** (e.g., coolant bottle **3000**), and various panels of outer body **80** (e.g., rear fenders, rear close-off panels, rear bumper, etc.). Illustratively, a forward portion of support area **2200** supports and may be coupled to an airbox **4030** of air intake assembly **2608** such that airbox **4030** is supported on at least tubes **2208**. Additionally, a rearward portion of support area **2200** supports and may be coupled to a muffler or silencer **4032** of exhaust assembly **2986**. As shown in FIG. **66**, brackets **4034** extend from at least tubes **2208** and are coupled to support arms **4036** which are attached to muffler **4032**.

(196) Referring to FIGS. **69** and **70**, front suspension **2040** is shown. As noted herein with respect to FIGS. **12** and **13**, front suspension **2040** includes upper alignment arm or A-arm **2280**, lower alignment arm or A-arm **2282**, and LFE or shock absorber **2284**. LFE **2284** is operably coupled to

lower A-arm **2282** through a clevis **2298**. More particularly, clevis **2298** has a generally U-shaped body extending between an upper end **4040** and a lower end **4042**. Lower end **4042** is removably coupled to lower A-arm **2282** with a fastener **4044**, such as a bolt. Clevis **2298** is configured to receive or straddle a portion of half shaft **2290** such that clevis **2298** and LFE **2284** do not interfere with the desired location or rotational movement of half shaft **2290**. In this way, half shaft **2290** is positioned vertically intermediate upper and lower ends **4040**, **4042** of clevis **2298**.

(197) As shown best in FIG. **70**, upper end **4040** of clevis **2298** is coupled to a lower portion of LFE **2284**. More particularly, upper end **4040** of clevis **2298** includes an aperture **4049** configured to receive a threaded end **4048** of LFE **2284**. Threaded end **4048** is aligned with a piston assembly **4046** of LFE **2284** and, illustratively, may be colinear with a rod portion of piston assembly **4046**. Additionally, the lower portion of LFE **2284** includes a spring retainer **4047** which is rotationally oriented or clocked through an interface to clevis **2298**. Illustratively, clevis **2298** includes a tab **2299** which fits within a slot **4045** on spring retainer **4047** to allow rod **4046** to pass through when installing spring retainer **4047** on LFE **2284**.

(198) By using clevis **2298**, front suspension **2040** and the steering assembly of vehicle **2** may be compactly packaged and a longer LFE **2284** may be used without compromising the weight of vehicle **2**.

(199) Referring to FIGS. **69-73**, front braking system **2420** is shown positioned adjacent front wheel hub **2292**. As shown, braking system **2420** is coupled to steering spindle **2390** and comprises brake caliper **2422**, disc pads **2424**, and brake disc **2426**. Hub **2292** includes a plurality of apertures **2430**, **2440** which receive fasteners (not shown) therethrough and disc **2426** includes apertures **2436** such that hub **2292** and disc **2426** may be coupled together and coupled to a portion of spindle **2390**. Caliper **2422** may include castellated pistons which to reduce weight, as disclosed further herein, and may facilitate thermal management of front braking system **2420**.

(200) Caliper **2422** and disc pads **2424** couple to a brake mount at **2442** (FIG. **72**) such that disc pads **2424** are held on opposite sides of brake disc **2426**. Alternatively, brake disc **2426** could be shown as a single component, comprising disc **426**, at least a portion of hub **2292**, and fasteners (not shown).

(201) The configuration of front braking system **2420** allows for integration of caliper **2422** with spindle **2390**, thereby reducing weight and size at ground-engaging members **4** and minimizing the need for additional protective measures for brake caliper **2422**. More particularly, braking system **2420** and the connection of steering arms **400** (FIGS. **12-15**) to spindle **2390** at ball joint connection **4041** (FIG. **71**) are positioned longitudinally rearward of the wheel's rotational axis **R** (FIG. **69**). As such, brake caliper **2422** may require less protection from debris and mud accumulation than in other positions and the weight and size of ground-engaging member **4** and/or front braking system **2420** may be reduced.

(202) Referring to FIG. **74**, a rear braking system **4050** is shown and may be similar to front braking system **2420**. Rear braking system **4050** includes brake caliper **4052**, disc pads (not shown), and brake disc **4054**. Caliper **4052** and the disc pads may be coupled to a brake mount at **4054** such that the disc pads are held on opposite sides of brake disc **4054**.

(203) Referring now to FIGS. **74-78**, rear suspension **2042** is shown. Rear suspension **2042** includes trailing arms **2500**, upper radius arms or rods **2506**, lower radius arms or rods **4062**, LFEs or shock absorbers **4060**, and toe links or arms **4064**. As shown best in FIG. **74**, LFE **4060** is coupled to an upper surface of trailing arm **2500** through a shock mount **4066**. Shock mount **4066** is positioned longitudinally along the length of trailing arm **2500** and is positioned longitudinally forward of a rear end portion or knuckle carrier **4058** of trailing arm **2500**. Knuckle carrier **4058** may be stamped to reduce the weight of rear suspension **2042**. Knuckle carrier **4058** may be cast to reduce the complexity of rear suspension **2042**. The upper surface of trailing arm **2500** also may include a mount **4067** configured to receive a portion of torsion bar assembly **510** (FIG. **21**).

(204) Knuckle carrier **4058** of trailing arm **2500** includes an opening **4068** configured to receive

rear half shaft **2078**. Knuckle carrier **4058** further includes a rearward surface **4070** configured to support outer ends of upper and lower radius rods **2506**, **4062**. More particularly, rearward surface **4070** includes a mounting member **4072** configured to support radius rods **2506**, **4062**. In one embodiment, mounting member **4072** is integrally formed with knuckle carrier **4058** of trailing arm **2500**, while in other embodiments, mounting member **4072** is removably coupled to knuckle carrier **4058**. The outer ends of radius rods **2506**, **4062** are positioned longitudinally intermediate mounting member **4072** and knuckle carrier **4058** such that mounting member **4072** defines the rearwardmost surface of rear suspension **2042**. Illustratively, radius rods **2506**, **4062** are positioned longitudinally rearward of rear half shaft **2078**.

(205) Referring still to FIGS. **74-78**, trailing arm **2500** includes an opening **4074** which extends between an inner portion **4076** and an outer portion **4078** of trailing arm **2500**. Inner and outer portions **4076**, **4078** may define a clamshell design and are stamped components which join together to define trailing arm **2500**. Portions **4076**, **4078** may be symmetrical on the right and left sides of rear suspension **2042** which minimizes tooling during the manufacturing process, thereby reducing cost. Additionally, because portions **4076**, **4078** are stamped, rear suspension **2042** may be made lighter.

(206) Trailing arm **2500** extends between a forward portion **4090** and knuckle carrier **4058**. Forward portion **4090** includes a coupler **4092** configured to operably coupled to frame **2020** and allow for generally vertical movement of trailing arm **2500** during operation of vehicle **2**. Knuckle carrier **4058** is coupled to a rearward portion **4091** of trailing arm **2500** which, illustratively, generally defines a “V” shape at **4093**. Rearward portion **4091** may be fixed to knuckle carrier **4058** through welding. The configuration of trailing arm **2500** and the connection to knuckle carrier **4058** may distribute loads at trailing arm **3500** more efficiently and increase the weld content at the interface between rearward portion **4091** and knuckle carrier **4058**.

(207) Opening **4074** extends completely through trailing arm **2500** and is configured to receive a portion of toe link **4064**. More particularly, toe link **4064** extends generally longitudinally between a forward end **4080** and a rearward end **4082**. Forward end **4080** includes a coupler **4084** configured to operably couple with a portion of frame **2020**. Rearward end **4082** includes a coupler **4086** configured to be operably coupled to a hub assembly **4088** of rear ground-engaging member **6** (FIG. **1**). Coupler **4086** may define a joint loaded in double shear. A corresponding bearing may be loaded radially and, as such, the risk of coupler **4086** and the bearing pulling apart if coupler **4086** becomes worn is minimized.

(208) Coupler **4086** may be positioned within an opening **4087** of knuckle carrier **4058** of trailing arm **2500**. As shown best in FIGS. **76A-76C**, coupler **4086** may include a bolt **4400**, an eccentric washer **4402**, and a nut **4404** which, collectively, provide the ability to adjust alignment of toe link **4064** by rotating bolt **4400** before tightening nut **4404**. Bolt **4400** may be mechanically coupled to washer **4402**, through knurling, a D-profile, or other shape. The washer profile is eccentric to the bolt axis and knuckle carrier **4058** may have locating tabs that washer **4402** contacts. In this way, as bolt **4400** is turned, washer **4402** moves the bolt axis fore and aft in the opening. Once bolt **4400** is rotated to provide the desired alignment of toe link **4064**, the bolt head is fixed in position and nut **4404** is tightened to the desired torque.

(209) As shown best in FIG. **75**, toe link **4064** extends diagonally through opening **4074** such that forward end **4080** is positioned inwardly of outer portion **4078** of trailing arm **2500** while rearward end **4082** is positioned outwardly of inner portion **4076** of trailing arm **2500**. In this way, the configuration of rear suspension **2042** may be compact so as to not interfere with other components or systems of vehicle **2**, such as the powertrain or driveline, but opening **4074** provides sufficient space for toe link **4064** to move with or relative to trailing arm **2500** when needed.

(210) Additional details of rear suspension **2042** may be disclosed in U.S. patent application Ser. No. 16/266,797, filed Dec. 20, 2018, and entitled “REAR SUSPENSION ASSEMBLY FOR A VEHICLE”, the complete disclosure of which is expressly incorporated by reference herein.

(211) Referring to FIGS. 79-93, portions of the powertrain of vehicle 2 are disclosed herein, the powertrain of vehicle 2 includes at least engine 50 and CVT 52 (FIGS. 23 and 24). At the interface between engine 50 and CVT 52 may be an arc spring assembly 4100 (also shown in FIG. 41). More particularly, arc spring assembly 4100 may be positioned intermediate a crankcase of engine 50 and drive clutch 978 (FIGS. 39-41) of CVT 52 and, in one embodiment, directly coupled to a crankshaft 4102 of engine 50 and drive clutch 978 of CVT 52. Illustratively, as shown best in FIG. 81, arc spring assembly 4100 is directly coupled to an input end 4104 of crankshaft 4102. By positioning arc spring assembly 4100 at input end 4104 of crankshaft 4102 and adjacent drive clutch 978 of CVT 52, vibrations at engine 50 are absorbed and do not transfer to CVT 52. In other words, arc spring assembly 4100 reduces or dampens the applied shaking forces from crankshaft 4102 during the engine firing events, thereby increasing the life of CVT 52 by reducing the torsional pulsation to decrease the temperature of the CVT belt.

(212) Arc spring assembly 4100 includes an arc spring damper assembly 4110, a ring gear 4112 positioned around a flex plate 4111, and a hub 4114. Hub 4114 is coupled to input end 4104 of crankshaft 4102. Hub 4114 includes apertures 4120, at least some of which align with apertures 4122 on input end 4104 to receive fasteners 4118 (e.g., bolts, rivets, etc.). Flex plate 4111 and ring gear 4112 are positioned adjacent hub 4114 such that hub 4114 is positioned intermediate flex plate 4111 and input end 4104 of crankshaft 4102. Ring gear 4112 has a larger diameter than hub 4114 and includes a plurality of teeth 4124 along an outer circumference thereof. Flex plate 4111 includes a plurality of locating members (e.g., dowels or pins) 4113 configured to properly locate CVT 52 relative to engine 50 (e.g., may be used for proper positioning on the flywheel of engine 50). Additionally, flex plate 4111 includes a plurality of mounting members 4115 (e.g., studs) configured to couple to arc spring damper assembly 4110. Flex plate 4111 also includes a plurality of apertures 4126 configured to align with apertures 4120 and 4122 to receive fasteners 4118 for coupling flex plate 4111 and hub 4114 to input end 4104 of crankshaft 4102.

(213) A mounting plate 4116 may be positioned on the opposite side of flex plate 4111 relative to hub 4114 such that flex plate 4111 is positioned intermediate mounting plate 4116 and hub 4114. Mounting plate 4116 includes a plurality of apertures 4128 configured to align with apertures 4120, 4122, 4126 to receive fasteners 4118 for coupling mounting plate 4116 with flex plate 4111, hub 4114, and input end 4104.

(214) Arc spring damper assembly 4110 is positioned adjacent mounting plate 4116 such that mounting plate 4116 is intermediate flex plate 4111 and arc spring damper assembly 4110. Arc spring damper assembly 4110 includes a spring body cover 4130, a drive plate 4132, and a threaded aperture 4134. Threaded aperture 4134 is configured to receive a portion of shaft 824 and/or shaft 828 (FIG. 41) such that crankshaft 4102 drives drive clutch 978 of CVT 52 (FIG. 41) through arc spring assembly 4100. Threaded aperture 4134 may be centrally positioned on spring body cover 4130. In this way, arc spring assembly 4100 is directly coupled to crankshaft 4102 and drive clutch 978. Through this direct connection between crankshaft 4102 and drive clutch 978, arc spring assembly 4100 becomes the decoupling device between drive clutch 978 and engine 50. The use of arc spring assembly 4100 reduces side load on crankshaft 4102 to increase the life of engine 50 and the life of the belt of CVT 52, as further disclosed herein.

(215) Drive plate 4132 includes couplers 4136 which are configured to receive mounting members 4115 on flex plate 4111 for coupling together arc spring damper assembly 4110 and flex plate 4111.

(216) As shown best in FIGS. 83 and 84, spring body cover 4130 includes at least one spring 4138 extending at least partially around threaded aperture 4134 and having a coiled shape. Springs 4138 may be retained by a spring retainer 4139. Springs 4138 are configured to receive and absorb vibrational forces from engine 50 to minimize transfer of such vibrational forces to CVT 52, as disclosed further herein. In this way, arc spring assembly 4100 is configured to increase the life of CVT 52 by minimizing wear thereto.

(217) More particularly, the torque path through arc spring damper assembly 4110 is shown as the

arrows in FIG. 84. Torque initially is transmitted through drive plate **4132** and then transmitted through spring body cover **4130**. Torque is transmitted through spring body cover **4130** and then transmits along fasteners **4118** between spring body cover **4130** and spring retainer **4139**. Torque then transmits through spring retainer **4139** before transmitting through springs **4138** and ultimately transmitting through a flange **4131** and a hub **4137** generally surrounding threaded opening **4134**. Springs **4138** provide the damper capacity stop when the inner diameter coils go solid. It may be appreciated that if the torque increases to an amount greater than the spring capacity, the torque still follows the same torque path disclosed in FIG. 84.

(218) Referring to FIGS. **85-86B**, a further portion of the powertrain of vehicle **2** is disclosed. Illustratively, and similar to that disclosed in FIG. **53** with respect to supporting front drive **70** on frame **20** or **2020**, lower longitudinal tube portions **1330** include threaded inserts **1340** whereas upper longitudinal tube portions **1332** include threaded inserts **1342**. Front pedestal plate **1334** is coupled to the lower longitudinal tube portions **1340** and upper longitudinal tube portions **1342** and includes an opening **1350** generally conforming to allow the passage of front drive **70** therein, as described in greater detail herein. Side plate **1336** is generally coupled to the left-hand side lower longitudinal tube portion **1330** and upper longitudinal tube portion **1332**. Left plate portion **1336** includes an opening at **1360** to allow access between front drive **70** and left front wheel. Left side plate **1336** further includes a front opening at **1362** and a rearward opening at **1364** which align with mounting inserts **496**, as further described herein. A front pedestal **1370** is positioned below aperture **1362** and a rearward pedestal **1372** is positioned below opening **1364**. Pedestal **1370** defines a mounting surface at **1374** while pedestal **1372** defines a mounting surface at **1376**.

(219) Fasteners **1410**, **1412** are installed through plate portions **1336** and through front drive **70** and are tightened without any preload loss at the joint. The plate portions **1336** are castings which have smooth radii and offer complete contact. More particularly, the boss on plate portions **1336** has proper contact with an isolator to prevent the isolator from tearing and cutting. Once fasteners **1410**, **1412**, which are horizontal fasteners, are tightened, vertical fasteners **1400**, **1402** are tightened. If the order of tightening the horizontal and vertical fasteners is reversed, then plate portions **1336** will be pre-stressed in bending and may fatigue early. In this way, front drive **70** utilizes a slip joint to remove the bolt preload and facilitate installation.

(220) Referring to FIGS. **87A** and **87B**, and similar to front drive **70**, rear drive **2076** may be similarly mounted to frame **2020**. More particularly, rear drive **2076** may include bosses **4410** extending generally laterally and configured to receive removable fasteners **4412** therethrough. Fasteners **4412** also extend through openings in a mounting bracket **4414** and, as such, couple together mounting bracket **4414** and the housing of rear drive **2076** with couplers **4416**. It may be appreciated that brackets **4414** are removable from frame **2020** to facilitate service and assembly of rear drive **2076**.

(221) Brackets **4414** are coupled to a brace **4420** extending between and coupled to frame tubes **2174**. More particularly, fasteners **4422** extend generally vertically between corresponding apertures in brace **440** and brackets **4414** to couple together brackets **4414** and brace **4420**. Fasteners **4422** extend through isolators **4421** on bracket **4414**. Couplers **4424** are configured to receive a portion of fasteners **4422** to secure bracket **4414** to brace **4420**. Because rear drive **2076** is coupled to brackets **4414**, rear drive **2076** also is coupled to brace **4420** through brackets **4414**.

(222) The housing of rear drive **2076** further includes second bosses **4418** configured to receive fasteners **4426** in order to couple a forward portion of rear drive **2076** to upstanding brackets **4428**. Illustratively, fasteners **4426** extend through apertures in upstanding brackets **4428** and through bosses **4418** of rear drive **2076** in order to couple with couplers **4430**. Upstanding brackets **4428** extend upwardly from frame tubes **2172**.

(223) Referring still to FIGS. **87A** and **87B**, rear drive **2076** may be mounted in double shear and also mounted in a manner to avoid clamp load loss when the installation sequence is followed. To avoid clamp load loss, the following installation sequence may be used. First, rear drive **2076** is

assembled with frame **2020** by dropping rear drive **2076** down into the rear envelope between upstanding brackets **4428**. Although upstanding brackets **4428** may be welded to frame tubes **2172**, upstanding brackets **2172** create a cradle to hold rear drive **2076** in place while fasteners **4426** are then assembled through bosses **4418**. Next, brackets **4414** are installed in the fore-aft direction into the envelope of brace **4420** and fasteners **4422** are then installed. Then, vertical fasteners **4422**, which may be double shear bolts, are added loosely. Fasteners **4426** are tightened first, then fasteners **441** are tightened. Lastly, fasteners **4422** are tightened. Accordingly, fasteners **4412** have no preload loss in the joint and allow some flexibility and tolerance stack up with this configuration.

(224) Referring to FIGS. **88-93**, the powertrain of vehicle **2** includes a breather tube or conduit **4140**. Breather tube **4140** is configured to prevent engine oil from entering the combustion system of engine **50** if vehicle **2** is in a tip- or roll-over situation. For example, at approximately 80° lean relative to vertical, sump oil for engine **50** may move through breather tube **4140** and could enter the intake manifold of engine **50**, which may cause damage to the cylinders and other components of engine **50**. As such, there is a need for breather tube **4140** to mechanically close off this flow path to the intake manifold during a roll-over situation.

(225) Breather tube **4140** includes a first or engine portion **4142** configured to be coupled with main oil sump **648** of engine **50** (FIG. **25**) and a second or intake portion **4144** configured to be coupled with an intake manifold of engine **50**. More particularly, first portion **4142** includes a coupler or connector **4146** configured as a quick-connect coupler for attaching to sump **648** or other portion of engine **50** and second portion **4144** includes a coupler or connect **4148** configured as a quick-connect coupler for attaching to the intake manifold.

(226) Additionally, breather tube **4140** includes a check valve **4150** positioned at an interface **4152** of first and second portions **4142**, **4144**. Check valve **4150** includes a housing **4154** having a first end **4156** and a second end **4158** and a check ball **4153**. First end **4156** is positioned generally in first portion **4142** of breather tube **4140** while second end **4158** is positioned generally in second portion **4144** of breather tube **4140**. Housing **4154** may be a single component configured to house check ball **4153** such that first and second ends **4156**, **4158** may be integrally formed together. In various embodiments, housing **4154** may be comprised of aluminum tubing or injection-molded plastic which is ultrasonically welded together. The materials comprising housing **4154** and check ball **4153** are configured to withstand high engine oil temperatures (e.g., approximately 150° C.) and completely seal hot oil from entering the intake manifold of engine **50**.

(227) First end **4156** includes at least one stop feature **4160**, configured as an indentation or reduced diameter portion of housing **4154**, which prevents movement of check ball **4153** past this portion of first end **4156**. Second end **4158** includes a seat **4162** configured to receive check ball **4153** when check ball **4153** is within second end **4158**. Seat **4162**, which may be comprised of a rubber material, includes an orifice **4164** extending therethrough and is configured to receive a fluid depending on the operation and position of vehicle **2**. In one embodiment, the flow area through housing **4154** may be approximately seven times larger than the flow area through orifice **4164**. Both first and second ends **4156**, **4158** include guides **4166** configured to ensure that check ball **4153** remains centered within housing **4154** for complete sealing, if needed.

(228) In operation, and referring to FIGS. **92** and **93**, if vehicle **2** encounters a tip- or roll-over situation, check ball **4153** moves towards seat **4162** to prevent oil from sump **648** flowing into the intake manifold of engine **50** (FIG. **92**). However, when vehicle **2** is upright and operating along a ground surface, check ball **4153** may move away from seat **4162** in order to allow gas pressure from the crankcase of engine **50** or other fluids to vent during operation of the powertrain (FIG. **93**).

(229) Referring to FIGS. **94-96**, the powertrain of vehicle **2** includes air intake system **608**, **2608**, as disclosed herein. Air intake system **608**, **2608** includes intake ports **4170** positioned generally adjacent utility bed **2088**. Illustratively, intake ports **4170** are positioned on right and left sides of

vehicle 2 and one intake port **4170** is configured to provide air to engine **50** and the other intake port **4170** is configured to provide air to CVT **52**. The incoming air from intake port **4170** to engine **50** is provided for combustion while the incoming air from the other intake port **4170** to CVT **52** provides cooling air for cooling clutches **978**, **980** and the belt.

(230) Intake ports **4170** each include a filter or other cover **4172** coupled to a bezel **4174**. Bezel **4174** may be defined as part of outer body **80**, **2080** and may receive filter **4172** (e.g., be removably coupled with filter **4172**) or may be integrally formed with filter **4172**. Ports **4170** face outwardly from vehicle **2** such that ports **4170** face the ambient air surrounding vehicle **2** and are configured to pull ambient air therein.

(231) To minimize noise caused by the incoming air flowing into ports **4170**, bezel covers or blockers **4176** may be positioned along a portion of bezels **4174**. More particularly, because ports **4170** are positioned generally at the same vertical height as a portion of the seat back in seating area **22**, noise at ports **4170** may be heard by the operator and/or passenger; however, bezel covers **4176** deflect sound rearwardly and away from seating area **22**.

(232) Illustratively, bezel covers **4176** include an inwardly-facing wall **4176a**, an upper surface **4176b**, and a rearwardly-extending portion **4176c**. In this way, bezel covers **4176** generally surround upper and inner surfaces of bezels **4174** while leaving the outer surface of bezel **4174** and filter **4172** fully exposed for receiving the ambient air. Bezel cover **4176** may be removably coupled to bezel **4174** through fasteners **4178** extending through at least upper surface **4176b**. In this way, bezel covers **4176** may be removed for cleaning and removing mud or dirt accumulations thereon. It may be appreciated that bezel covers **4176** do not extend into utility bed **2088** and, therefore, do not impinge on the available cargo volume or space of utility bed **2088**.

(233) Referring to FIGS. **97-99**, the powertrain of vehicle **2** includes exhaust **986**, **2986**, as disclosed herein. Exhaust **986**, **2986** includes muffler **4032** supported at the rear end of vehicle **2** and, more particularly, supported by support area **2200**. Exhaust **986**, **2986** further includes an exhaust outlet or tailpipe **4180** comprised of a conduit portion **4182** and an outlet portion **4184**. Conduit portion **4182** may be removably coupled or integrally formed with a portion of muffler **4032** and outlet portion **4184** is fluidly coupled to muffler **4032** through conduit portion **4182**.

(234) Illustratively, outlet portion **4184** is removably coupled to conduit portion **4182** at an intersection or coupling location **4188** and is retained on conduit portion **4182** with at least a removable fastener **4186** (FIG. **98**). A collar or other coupling or retaining member also may be used to join outlet portion **4184** to conduit portion **4182** at intersection **4188**.

(235) Outlet portion **4184** may include a plurality of sides which intersect with each other. Illustratively, the shape of outlet portion **4184** is defined by four sides and generally defines a trapezoid. As shown best in FIGS. **97** and **98**, a length of an upper side **4184a** may be greater than that of a lower side **4184b**. Left and right sides **4184c**, **4184d** may extend between upper and lower sides **4184a**, **4184b** and are coupled thereto such that left and right sides **4184c**, **4184d** extend diagonally inwards towards centerline **L** to couple with lower side **4184b**.

(236) Outlet portion **4184** is defined as a double-wall outlet which includes a first wall segment **4190** extending from and generally colinear with conduit portion **4182** and a second wall segment **4192** surrounding a portion of first wall segment **4190**. In one embodiment, first and second wall segments **4190**, **4192** are integrally formed with each other and, because of that, second wall segment **4192** increases the return edge of outlet portion **4184**. In this way, first and second wall segments **4190**, **4192** radially overlap each other to define the double-wall structure. In various embodiments, any portion of exhaust **986**, **2986** may include an insulated cover or wrapping to shield adjacent components of vehicle **2** from the heat produced within exhaust **986**, **2986** and/or to strategically manage the heat within exhaust **986**, **2986** at various points along the system.

(237) In one embodiment, exhaust **2986** is a center exhaust such that outlet portion **4184** is positioned along centerline **L** (FIG. **5**). However, various panels of outer body **80** also may be positioned at this location and, therefore, it is necessary to ensure that the temperature of exhaust

**2986** at the location of the body panels of outer body **80** does not exceed a threshold which is detrimental to the body panels. By configuring outlet portion **4184** as a double-wall outlet, the cumulative thermal conductivity of outlet portion **4184** decreases, thereby protecting the body panels adjacent outlet portion **4184**.

(238) Referring to FIG. **100**, a cooling assembly may be provided with the powertrain of vehicle **2** in order to provide cooling to the various components of the powertrain, such as engine **50**. The cooling assembly includes coolant bottle **3000**, which may be similar to coolant bottle **1000** of FIG. **42**. Coolant bottle **3000** may be supported on support area **2200**, as shown in FIG. **66**, and, illustratively, may be supported on tubes **2208** at a position generally adjacent airbox **4030**. Coolant bottle **3000** includes a housing **3002** configured to store coolant fluid and, more particularly, includes an upper housing portion **3002a** and a lower housing portion **3002b**. Lower housing portion **3002b** includes a notched or recessed portion **3003** positioned adjacent a lower or bottom surface **3004** of housing **3002**.

(239) Recessed portion **3003** is configured to receive a sensor **3006**. Sensor **3006** is configured to detect the coolant level within coolant bottle **3000**. Based on the location of recessed portion **3003** and sensor **3006** (i.e., the location adjacent lower surface **3004**), sensor **3006** can detect when the coolant level in housing **3002** is low. Sensor **3006** is electrically or otherwise operably coupled to a control system of vehicle **2** and, therefore, if a controller receives an input from sensor **3006** indicating that the coolant level in coolant bottle **3000** is low, the control system may provide a visual or auditory alert to the operator. If necessary, the control system may control vehicle operation and/or parameters of the powertrain based on the coolant level within coolant bottle **3000** and the output of sensor **3006**.

(240) Referring to FIG. **101**, the powertrain of vehicle includes fuel tank **3050** similar to fuel tank **1050** disclosed in FIGS. **44** and **45**. Fuel tank **3050** may include a vent conduit **4194** positioned along a portion of frame **2020** and directed towards the rear of vehicle **2**. More particularly, vent conduit **4194** extends upwardly towards frame member **2182** and along frame tube **2174** to enter frame tube **2174** at an opening **4196** (also shown in FIGS. **56** and **63-65**). Frame tube **2174** is a hollow frame tube which also includes an opening **4198** (FIGS. **57** and **63-65**) at a rear end thereof. Vent conduit **4194** may extend through opening **4196** and along a length of frame tube **2174** towards opening **4198**. In this way, fuel (e.g., vapor) that is vented from fuel tank **3050** flows through vent conduit **4194** and into frame tube **2174** at opening **4196**. The fuel that vents from conduit **4194** into frame tube **2174** then flows rearwardly through at least a length of vent conduit **4194** and/or frame tube **2174** and exits frame tube **2174** and vehicle **2** at opening **4198**. As such, vented fuel is directed towards right side of vehicle **2** at frame tube **2174** and is directed away from heat-producing components (e.g., of the powertrain), even when vehicle **2** is traversing on a hill or is leaning, for example, to the left side. In other words, the configuration of frame tube **2174**, openings **4196**, **4198**, and vent conduit **4194** ensures that fuel does deviate from the path within frame tube **2174** and flow towards a hot area of frame **2020** when vehicle **2** is at certain orientations.

(241) Referring to FIGS. **102** and **103**, in operation, the powertrain of vehicle **2** may be configured for pull-pull shifting. The powertrain of vehicle **2** further includes shiftable transmission **2056** which may be similar to shiftable transmission **56** of FIG. **3**. Shiftable transmission **2056** is operable by a shifter assembly **2060** which may be similar to shifter assembly **60** of FIG. **5**. Illustratively, shifter assembly **2060** is operably coupled to a first cable **4200** and a second cable **4202** both of which extend between shifter assembly **2060** and shiftable transmission **2056**.

(242) Cables **4200**, **4202** are operably coupled to a pulley **4204** of shiftable transmission **2056**. At least a portion of cables **4200**, **4202** is supported on a bracket **4206** extending from a housing **4208** of shiftable transmission **2056**. Bracket **4206** is removably coupled to housing **4208** with fasteners **4209**. Each cable **4200**, **4202** is coupled to pulley **4204** at a connection location and, as shown best in FIG. **103**, cable **4200** is coupled to pulley **4204** at connection location **4210** and cable **4202** is

coupled to pulley **4204** at connection location **4212**. Connection locations **4210**, **4212** are fixed points on pulley **4204** such that movement of cables **4200**, **4202** pulls pulley **4204** at the corresponding connection location **4210**, **4212** to cause a shift change in shiftable transmission **2056**. Pulley **4204** keeps the effective radius of cables **4200**, **4202** constant and, therefore, the shift force also stays constant. By using pulley **4204** and the two pull cables **4200**, **4202**, the force required to shift remains constant.

(243) Cables **4200**, **4202** are configured as pull cables such that movement in cable **4200**, **4202** pulls on pulley **4204** based on movement of shifter assembly **2060**, compared to known designs which require a combination of pulling and pushing movements on the transmission bell crank to shift the transmission. By configuring cables **4200**, **4202** as pull cables, the bell crank of traditional shiftable transmissions may be eliminated.

(244) In one embodiment, cables **4200**, **4202** are different sizes to ensure proper installation on vehicle **2**. More particularly, connection locations **4210**, **4212** of pulley **4204** are defined as openings with different diameters to corresponding to the different diameters of cables **4200**, **4202** such that only cable **4200** may be coupled to pulley **4204** at connection location **4210** and cable **4202** may be coupled to pulley **4204** at connection location **4212**. Similarly, shifter assembly **2060** also includes differently-sized openings corresponding to each of cables **4200**, **4202**. Further, bracket **4206** also may include openings having different diameters such that bracket **4206** receives cable **4200** through a first opening for coupling with connection location **4210** and bracket **4206** receives cable **4202** through a second opening (having a different diameter than that of the first opening) for coupling with connection location **4212**.

(245) In operation, when shifter assembly **2060** is moved to a particular position by the operator to indicate the desired gear, either cable **4200** or cable **4202** moves in response to the movement of shifter assembly **2060**. The movement of cable **4200** or cable **4202** pulls on pulley **4204**. The movement of pulley **4204** causes a shift change in shiftable transmission **2056**. It may be appreciated that because cables **4200**, **4202** are not connected to each other (e.g., because each cable **4200**, **4202** has a separate connection location **4210**, **4212** on pulley **4204**), when one cable moves, there is no corresponding and opposite movement of the other cable. Instead, both cables **4200**, **4202** are configured for a pulling motion on pulley **4204** based on the movement of shifter assembly **2060**.

(246) Referring to FIGS. **104** and **105**, in order to minimize noise and vibrations from the powertrain and/or other components or systems of vehicle **2** within seating area **22**, vehicle **2** may include panels **4220**. Panels **4220** may be configured as two-part panels which include a first portion **4222** and a second portion **4224**. First and second portions **4222**, **4224** may be coupled together in various ways, such as with adhesive or mechanical fasteners. First and second portions **4222**, **4224** are coupled to vehicle **2** through fasteners **4226**, **4228**. Fasteners **4226**, **4228** may be dowels, pins, bolts, screws, or other removable coupling or locating components.

(247) Illustratively, first portion **4222** is positioned forward of second portion **4224** such that first portion **4222** is exposed to seating area **22** while second portion **4224** is not. Fasteners **4226** extend through openings **4230** in first portion **4222** and fastener **4228** extends through an opening **4232** in first portion **4222** in order to couple with frame **2020** or other portions of vehicle **2** within or defining seating area **22**.

(248) First portion **4222** includes a curved or arcuate wall **4234** and a linear or flat wall **4236**. Similarly, second portion **4224** includes a curved or arcuate wall **4238** and a linear or flat wall **4239**. Walls **4234**, **4238** have complementary shapes and, therefore, wall **4234** is generally received within wall **4238**. Walls **4236**, **4239** also have complementary shapes and may be in contact with each other.

(249) First portion **4222** may be comprised of a polypropylene and fiberglass material, such as Azdel material, available from Federal Foam Technologies, Inc. Second portion **4224** may be comprised of a sound-dampening foam material. In this way, first and second portions **4222**, **4224**

absorb sound and vibrations from other portions of vehicle **2** so as to minimize noise and vibrations within seating area **22**. The materials of first and second portions **4222**, **4224** also may be used to reduce the weight of vehicle **2** compared to other sound-dampening or vibration-absorbing materials. Additionally, because at least first portion **4222** is exposed within seating area **22** and, therefore, is exposed to water, mud, dirt, and debris that may enter seating area **22**, panels **4220** may be configured as waterproof panels.

(250) Referring to FIGS. **106-111**, and as disclosed further herein with respect to outer body **80** (FIG. **1**), outer body **2080** of vehicle **2** may include various features. Hood **2082** of outer body **2080** may include a hinge assembly **4240** for coupling a grille **4242** to various surfaces of hood **2082**. Grille **4242** may be used to receive and direct ambient air into at least one conduit for components of the powertrain (e.g., an air intake duct of CVT **52**). Grille **4242** is positioned within an opening **4243** of hood **2082** and, illustratively, hood **2082** includes openings **4243** on both the left and right sides thereof to support two grilles **4242**. Grilles **4242** and the corresponding intake ports at grilles **4242** face laterally outward from centerline **L** (FIG. **5**) to receive ambient air for a powertrain component (e.g., CVT **52**).

(251) Hinge assembly **4240** includes an enclosure **4246** extending generally around a perimeter of grille **4242**. Enclosure **4246** is coupled to hood **2082** through fasteners **4248**, **4250**. More particularly, fasteners **4248** extend through openings **4252** within enclosure **4246** and receive fasteners **4250** for coupling enclosure **4246** (encompassing grille **4242**) to hood **2082**.

(252) Hinge assembly **4240** further includes a deflector panel **4254** coupled to grille **4242** through an upper panel **4256**. Upper panel **4256** extends generally inwardly and horizontally from an upper surface of grille **4242** and is positioned below a portion of hood **2082**. Deflector panel **4254** extends generally vertically and inwardly relative to upper panel **4256** and grille **4242** and is coupled to upper panel **4256** at a living hinge **4258**. The pattern of grille **4242** and the position of deflector panel **4254** relative to grille **4242** minimizes the likelihood that debris or water, which may inadvertently flow through grille **4242**, flows into the air intake conduits coupled to the powertrain. Instead, if debris or water flows through grille **4242** and under hood **2082**, the debris or water will contact an outer surface of deflector panel **4254** and will fall vertically downward and out of vehicle **2**, rather than flowing inwardly under hood **2082** and into the air intake conduit in fluid communication with grilles **4242**. More particularly, the pattern of grille **4242** prevents large debris from entering hood **2082** while deflector panel **4254** is inward of grille **4242** and blocks the line of sight to the conduit (e.g., CVT air intake conduit) from water spray, small debris, etc.

(253) Hinge assembly **4240** further includes a tab **4260** extending from deflector panel **4254**. Tab **4260** is coupled to deflector panel **4254** at a living hinge **4262**. Tab **4260** includes a slot or opening **4264** configured to receive a strap **4266** coupled to enclosure **4246** and/or grille **4242**. Strap **4266** at least partially extends through opening **4264** on tab **4260** and is secured thereto with a clip **4268**. In this way, deflector panel **4254** is coupled to grille **4242** through upper panel **4256**, tab **4260**, and strap **4266**.

(254) Deflector panel **4254** is further coupled to grille **4242** through an arm **4270** extending from grille **4242** and/or enclosure **4246** and extending through an opening **4272** of deflector panel **4254**. Arm **4270** may include a retaining feature **4274** configured to extend through opening **4272** and hook or otherwise retain arm **4270** on deflector panel **4254**.

(255) Referring now to FIGS. **112-114**, vehicle **2** may include various accessories. For example, vehicle **2** may include a lightbar assembly **4280** coupled to a portion of cab frame **2028**. Lightbar assembly **4280** may be used with or without a roof for vehicle **2**. In one embodiment, and where a roof is not included on vehicle **2**, lightbar assembly **4280** is coupled to a frame member **4282** extending generally between frame members **3072**. Illustratively, light bar assembly **4280** is coupled to gussets **4284** which are coupled to both frame members **3072** and **4282**. Gusset **4284** may have a generally a triangular shape which extends between frame members **3072** and **4282**. Gusset **4284** includes openings **4286** configured to receive fasteners **4288** for coupling lightbar

assembly **4280** to cab frame **2028**, as disclosed further herein. It may be appreciated that if a roof is included on vehicle **2**, the roof also may be mounted to cab frame **2028** at openings **4286** of gussets **4284** such that the roof and lightbar assembly **4280** share common mounting locations on cab frame **2028**.

(256) Lightbar assembly **4280** is comprised of at least a lightbar **4290**, mounting members **4292**, and wiring **4294**. Lightbar **4290** extends generally along the length of frame member **4282** and, in one embodiment, is approximately the same length as frame member **4282**. In alternative embodiments, lightbar **4290** has a length that is less than that of frame member **4282**. Lightbar **4290** includes end plates **4296** and a fastener **4298** extending therefrom. End plate **4296** and fastener **4298** are received within a recessed portion **4310** of mounting member **4292** and, more particularly, fastener **4298** extends through an aperture **4300** of mounting member **4292**. Fastener **4298** is coupled to a second fastener **4302** to secure lightbar **4290** to mounting member **4292**.

(257) Wiring **4294** is electrically coupled to lightbar **4290** and extends through a portion of mounting member **4292** and along a portion of frame member **3072** in order to couple with a power source of vehicle **2**.

(258) Mounting members **4292** may be diecast components comprised of at least a metallic material. Mounting member **4292** includes apertures **4304** which are configured to receive fasteners **4288** in order to support mounting member **4292** and, therefore, lightbar **4290** on cab frame **2028**. Apertures **4304** are positioned along a lower surface **4312** of mounting member **4292**. Lower surface **4312** further includes a slot or channel **4306** which is configured to receive and conceal a portion of wiring **4294**. An isolator **4308** may be positioned adjacent cab frame **2028** and mounting member **4294** to minimize vibrational transfer to lightbar **4290**.

(259) Referring now to FIGS. **115-126**, front suspension assembly **40** is shown in greater detail. As noted above, suspension assembly **40** includes upper alignment or A-arm **280** (FIG. **126**), lower alignment or A-arm **282** (FIGS. **115-116**), and linear force element (LFE) **284** which is shown as a spring over shock (FIG. **12**). Lower A-arm **282** includes a forward extension **5000** and a rearward extension **5002** which are respectively coupled to couplers **374**, **376** and **384**, **386** for mounting lower A-arm **282** to frame **20** (FIG. **8**). Forward and rearward extensions **5000**, **5002** generally extend laterally relative to the longitudinal centerline **L** (FIG. **5**). Forward and rearward extensions **5000**, **5002** are coupled together to define a laterally outer portion **5004** of lower A-arm **282**. Outer portion **5004** illustratively includes a ball joint housing **5006** which is configured to receive ball joint **394** (FIG. **13**). More particularly, outer portion **5004** includes a recess **5008** configured to receive a portion of ball joint housing **5006** such that ball joint housing **5006** extends outwardly from outer portion **5004** to define a laterally outermost extent of lower A-arm **282**. In one embodiment, ball joint housing **5006** may be a forged component.

(260) Referring still to FIGS. **115-120**, lower A-arm **282** may be comprised of a plurality of components. Illustratively, lower A-arm **282** is comprised of an upper stamping **5010**, a lower stamping **5012**, and an upper plate **5014**. Unlike a butted joint between stampings **5010**, **5012**, upper stamping **5010** is coupled to lower stamping **5012** at overlapping sides **5016** (shown best in FIG. **120**) such that upper stamping **5010** and lower stamping **5012** define a clamshell arrangement. In other words, a lower extent of upper stamping **5010** extends over and is in contact with an upper extent of lower stamping **5012** at overlapping sides **5016**. Overlapping sides **5016** extend generally the complete or full length of upper and lower stampings **5010**, **5012** and, illustratively, extend between ball joint housing **5006** and couplers **374**, **376**, **384**, **386**. Upper and lower stampings **5010**, **5012** may be permanently coupled together with welds, adhesive, or any other conventional coupling mechanisms.

(261) Lower stamping **5012** defines the lowermost surface of lower A-arm **282** while upper stamping **5010** defines at least a portion of the uppermost surface of lower A-arm **282**. More particularly, an upper surface **5018** (FIG. **115**) of upper stamping **5010** defines the upper surface of lower A-arm **282** along a length of lower A-arm **282** that extends approximately from upper plate

**5014** to couplers **374, 376, 384, 386**. However, as described further herein, upper surface **5018** of upper stamping **5010** at outer portion **5004** of lower A-arm **282** may be positioned vertically below or underneath at least a portion of upper plate **5014** such that upper plate **5014** defines the upper surface of lower A-arm **282** at outer portion **5004**.

(262) As shown in FIGS. **115-120**, upper plate **5014** is configured to be positioned over a portion of upper surface **5018** of upper stamping **5010** at outer portion **5004** and may be permanently coupled thereto with welds, adhesive, or any other conventional coupling mechanism. Upper plate **5014** extends approximately from a recess **5020** to legs **5022, 5024**. Recess **5020** is configured to generally align with recess **5008** of upper stamping **5010** and also receives ball joint housing **5006**. As shown best in FIG. **118**, a portion of upper stamping **5010** is exposed laterally outwardly of upper plate **5014** adjacent ball joint housing **5006**. As such, this tiered configuration of ball joint housing **5006**, upper stamping **5010**, and upper plate **5014** allows for a defined weld area which allows a weld to penetrate all three surfaces during a single welding operation.

(263) Legs **5022, 5024** extend along an upper surface of a portion of extensions **5000, 5002**, respectively. In this way, upper stamping **5010** and upper plate **5014** collectively define the uppermost surface of lower A-arm **282**, where upper stamping **5010** defines the uppermost surface at extensions **5000, 5002** and upper plate **5014** defines the uppermost surface at outer portion **5004**.

(264) Upper plate **5014** also includes side walls **5026** which extend along a portion of upper stamping **5010** and downwardly towards lower stamping **5012**. An upper surface **5028** of upper plate **5014** includes an opening **5030**, which will be described further herein.

(265) With continuing reference to FIGS. **115-120**, lower A-arm **282** further includes a suspension mounting assembly **5032**. Suspension mounting assembly **5032** is defined by a gusset plate **5034**, a spherical or bearing **5036**, and a retention member **5038**, illustrative a C-clip tension member. It may be appreciated that component **5036** is not a bushing, as may be conventionally used in other suspension arrangements, and, instead, is bearing because a bushing would constrain the motion of the shock.

(266) Gusset plate **5034** may be a forging or may be machined. Gusset plate **5034** has an opening **5035** (FIGS. **117A** and **117B**) configured to receive bearing **5036** therein. Bearing **5036** is partially retained within opening **5035** through a small tolerance or interference fit and, as such, by forming gusset plate **5034** through forging, opening **5035** is able to be sized to retain bearing **5036**. Retention member **5038** also is configured to retain bearing **5036** within opening **5035** of gusset plate **5035**. Bearing **5036** is configured to couple a lower end of LFE **284** to lower A-arm **282**, as is disclosed further herein.

(267) Gusset plate **5034** is configured to be received within and extend through opening **5030** of upper plate **5014**. More particularly, gusset plate **5034** is configured to be positioned atop a portion of upper surface **5018** of upper stamping **5010** at outer portion **5004** and extends through opening **5030** of upper plate **5014**. Because upper plate **5014** is coupled to both upper stamping **5010** and gusset plate **5034**, upper plate **5014** helps to join together upper stamping **5010** and gusset plate **5034**.

(268) Opening **5035** of gusset plate **5032** is exposed and configured to receive fasteners for coupling with LFE **284**. Gusset plate **5032** may be permanently coupled to upper stamping **5010** and/or upper plate **5014** with welds, adhesive, or any other conventional coupling mechanism. For example, at least a portion of a lower surface **5037** of gusset plate **5034** may be welded or otherwise permanently coupled to the upper surface of upper stamping **5010** while at least a portion of lateral surfaces **5039** of gusset plate **5034** may be welded or otherwise permanently coupled to upper plate **5014**.

(269) Referring now to FIGS. **121** and **122**, a lower end of LFE **284** is coupled to a connector, illustratively clevis **298**, and connector/clevis **298** is coupled to gusset plate **5034** to pivotably couple LFE **284** to lower A-arm **282**. More particularly, clevis **298** includes two clevis legs **350** which have openings **352** that are configured to align with opening **5040** of bearing **5036** to receive

a fastener **342** therethrough. A second fastener **344** is secured to fastener **342** such that fastener **342** extends through openings **352** on both clevis legs **350** and opening **5040** of bearing **5036** and is partially received within second fastener **344** to removably couple clevis **298**, and therefore the lower end of LFE **284**, to lower A-arm **282**.

(270) Clevis **350** also includes an upper portion **5042** which is coupled to both clevis legs **350** and positioned adjacent the lower end of LFE **284**. Upper portion **5042** of clevis **350** includes a recessed portion **5044** positioned above and adjacent an opening **5046** defined between clevis legs **350**. Opening **5046** is configured to receive half shafts **290** and recessed portion **5044** ensures that a predetermined minimum clearance **C** is maintained between half shaft **290** and upper portion **5042** of clevis **298** during full compression or rebound of suspension assembly **40**.

(271) More particularly, and referring to FIGS. **124** and **125**, FIG. **124** shows front suspension **40** at ride height (i.e., when stopped or during travel across generally flat terrain), where front suspension **40** is not at full rebound or full compression. In FIG. **124** it is apparent that clearance **C** between upper portion **5044** of clevis **298** and half shaft **290** is maintained and clevis **298** does not contact half shaft **290**. However, even when front suspension **40** is at full rebound or compression, as shown in FIG. **125**, clearance **C**, albeit smaller than the distance of clearance **C** of FIG. **124**, is maintained between upper portion **5044** of clevis **298** and half shaft **290**. In this way, clevis **298** does not make contact with half shaft **290** regardless of the position of front suspension **40**.

(272) To further maintain clearance **C**, gusset plate **5034** includes a stop surface **5048**. Stop surface **5048** is a ramped or angled surface which is angled inwardly towards legs **5022**, **5024** of upper plate **5014** is configured to allow clevis legs **350** to make contact before clevis **298** would contact half shaft **290**. In this way, stop surface **5048** of gusset plate **5034** limits the pivot motion of clevis **298** to prevent contact with half shaft **290**.

(273) Because LFE **284** is coupled to lower A-arm **282**, lower A-arm **282** experiences loads reacted by LFE **284**. More particularly, loads from LFE **284** travel through bearing **5036** and into spindle **390** (FIG. **12**) and, as such, are experienced by at least outer portion **5004** of lower A-arm **282**. In this way, by including upper plate **5014** with upper stamping **5010**, the stiffness of lower A-arm **282** is increased along a distance **5056** (FIGS. **115** and **119**) between ball joint housing **5006** (coupled to spindle **390** (FIG. **12**)) and suspension mounting assembly **5032** where the loads from LFE **284** are experienced. In other words, because upper plate **5014** spans a lateral distance adjacent ball joint housing **5006** to suspension mounting assembly **5032**, the major loads experienced at front ground-engaging members **4** and LFE **284** are more evenly distributed throughout lower A-arm **282** by upper plate **5014** and upper plate **5014** increases the stiffness at these locations of the major loads.

(274) Referring now to FIG. **126**, it may be appreciated that upper A-arm **280** has a similar configuration to that of lower A-arm **282** in that upper A-arm **280** includes an upper stamping **5050** and a lower stamping **5052** which overlap each other and are permanently coupled together at **5054**.

(275) While this invention has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

## Claims

1. A suspension arm for a vehicle, the suspension arm comprising: a first portion having an upper surface; and a second portion coupled to the first portion and positioned over at least a portion of the upper surface, and the first portion and the second portion together define an upper surface of the suspension arm, wherein the first portion and the second portion together define a location for

coupling with a shock absorber, wherein the first portion includes a first lateral extension and a second lateral extension, and the second portion extends along a portion of the first and second lateral extensions of the first portion.

2. The suspension arm of claim 1, wherein the second portion is coupled to an upper surface of the first portion.
3. The suspension arm of claim 1, wherein the second portion is coupled to a laterally outer portion of the first portion.
4. The suspension arm of claim 1, further comprising a ball joint housing and both the first and second portion have a length extending to the ball joint housing.
5. The suspension arm of claim 1, further comprising a mounting assembly for linear force element, and the second portion generally surrounds the mounting assembly.
6. The suspension arm of claim 1, further comprising a third portion coupled to the first portion.
7. The suspension arm of claim 6, wherein the third portion defines a lowermost surface of the suspension arm.
8. The suspension arm of claim 7, wherein the third portion has a height that overlaps a height of the first portion.
9. A suspension assembly for a vehicle, the suspension assembly comprising: an alignment arm including: a mounting assembly having an upper member; a bearing plate coupled to an upper surface of the upper member, the bearing plate including: a ball joint recess extending through a thickness of the bearing plate; and a bearing opening; a bearing member, and the bearing member is removably coupled within a portion of the bearing plate through the bearing opening; a ball joint housing on a laterally outer end of the alignment arm, wherein the ball joint recess is at the ball joint housing and reveals the ball joint housing; wherein a portion of the upper member is exposed by the ball joint recess, and the ball joint recess is configured to permit coupling of a ball joint with the portion of the upper member and a linear force element extending through the bearing opening, the linear force element movably coupled to the alignment arm at the bearing member, the linear force element configured to distribute a load along at least a portion the alignment arm.
10. The suspension assembly of claim 9, further comprising a connector extending between a lower end of the linear force element and the bearing member, and the connector is configured to receive a mechanical fastener, and the mechanical fastener extends through the bearing member.
11. The suspension assembly of claim 10, wherein the bearing plate includes a stop surface configured to limit rotation of the connector.
12. The suspension assembly of claim 9, wherein the alignment arm comprises a first upper portion, a second upper portion, and a lower portion coupled to at least the first upper portion, and the bearing plate is coupled to at least the first upper portion.
13. The suspension assembly of claim 12, wherein the bearing plate is coupled to the second portion.
14. The suspension assembly of claim 9, further comprising a clevis, and wherein the linear force element is removably coupled to a first end of the clevis, and the second end of the clevis is coupled to the bearing member.
15. The suspension assembly of claim 9, wherein the alignment arm defines a stamped portion including an upper stamping and an upper plate coupled to an upper surface of the upper stamping, wherein the load is distributed through at least a portion of the stamped portion.
16. The suspension assembly of claim 9, wherein the load is distributed along a lateral distance between the bearing and the ball joint housing.
17. A suspension arm for a vehicle, the suspension arm comprising: an upper portion having a lower extent extending downward relative to a longitudinal axis of the upper portion, the upper portion includes a ball joint housing; a lower portion having an upper extent extending upward relative to a longitudinal axis of the lower portion, the lower portion coupled to the upper portion such that the upper portion and the lower portion define a clamshell arrangement, and the upper

extent of the lower portion overlaps a lower extent of the upper portion to define overlapping sides of the upper and lower portions, and the upper portion is coupled to the lower portion at the overlapping sides; and a bearing plate coupled along the upper portion, the bearing plate extends along the longitudinal axis of the upper portion, and the bearing plate includes: a ball joint recess configured to reveal the ball joint housing; and a bearing opening spaced from the ball joint recess, the bearing opening configured to receive a linear force element.

18. The suspension arm of claim 17, wherein the upper and lower portions are stampings.

19. The suspension arm of claim 17, wherein the overlapping sides extend a complete length of both the upper and lower portions.

20. The suspension assembly of claim 17, further comprising a bearing plate coupled to an upper surface of the upper portion, wherein the upper portion includes a first lateral extension and a second lateral extension, and the bearing plate extends along a portion of the first and second lateral extensions of the first portion.

21. A method of assembling a suspension arm for a vehicle, the method comprising: providing a first portion; providing a second portion; providing a third portion; permanently coupling the first portion to the second portion along an upper surface of the first portion, wherein the first portion includes a first lateral extension and a second lateral extension, and the second portion extends along a portion of the first and second lateral extensions of the first portion, and wherein the first portion and the second portion define a coupling location for a shock absorber; aligning an upper extent of the third portion with a lower extent of the first portion; and permanently coupling the first portion to the third portion along the lower extent of the first portion.

22. The method of claim 21, wherein the first and second portions define an uppermost surface of the suspension arm.

23. The method of claim 21, further comprising: providing a suspension mount; and permanently coupling the suspension mount to the first and second portions.

24. The method of claim 23, further comprising positioning the suspension mount to extend through an opening of the second portion.

25. The method of claim 21, wherein providing a first portion includes stamping the first portion, and wherein providing a third portion includes stamping the third portion.

26. The method of claim 25, wherein permanently coupling the first portion to the second portion includes welding the first portion to the second portion, and wherein permanently coupling the first portion to the third portion includes welding the first portion to the third portion.

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